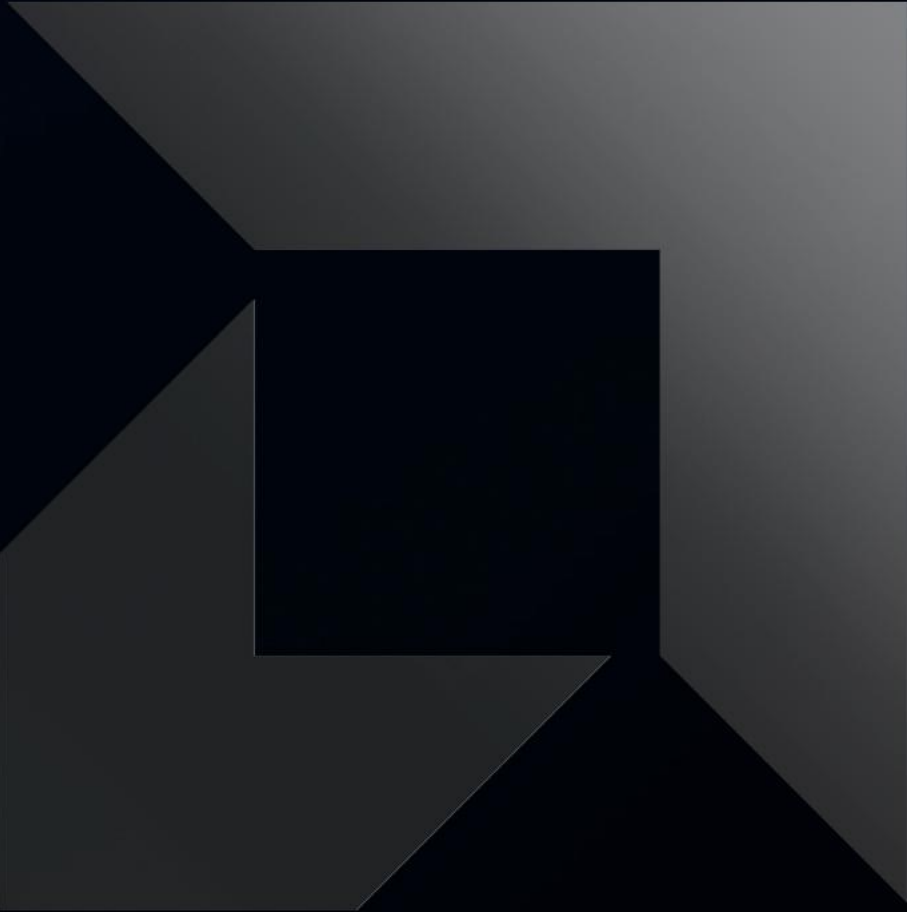




Reference Cluster Design Architecture MI300X / MI325X

1K GPU to 8K GPU

Version 4.2 – March 2026

Cluster - 1024 GPU to 8192 GPU (128 to 1024 Nodes) *

Cluster Size	1024 to 8192 GPU
Platforms	Dell XE9680 Lenovo SR685a V3 SMCI AS-8125GS
OS	Ubuntu 22.04 (or above)
Linux Kernel	5.15 – 6.80
ROCm	6.3.3 (or above)

*Genericized BoM – removing device specifics including dependencies

Storage Type	
Local Storage	1.6 TB (or greater)
Utility Storage	Pure, Vast, RYO
Bulk Storage	Pure, Vast, WekaIO
Scratch Storage	Vast, DDN, WekaIO, Hammerspace
Archive/Object Storage	S3 Compatible

Network - 1024 GPU to 8192 GPU (128 to 1024 Nodes) *

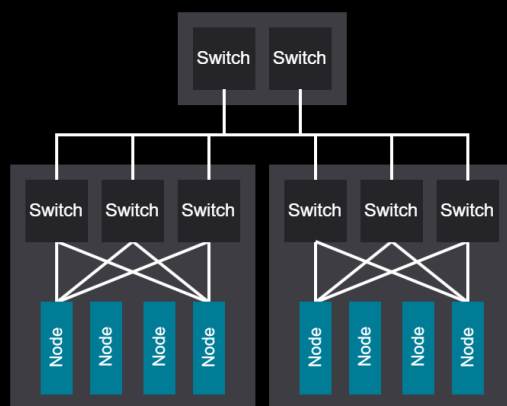
Backside Network Topology	2 Tier Rail Optimized / Fat Tree
NIC	Pollara 400 BCM957608 (Thor2)
Switch	Arista, Dell, Juniper, Cisco (TH 4/5, Scheduled fabrics), Nokia
Network OS	SONiC, Junos, EOS, IOS, DriveNets
Subscription Ratio	1:1.16=16% Undersubscribed (AMD recommendation)
Optics	Vendor ACL/HCL
Fabric	ROCEv2 Ethernet

Frontside Network Segment	Adapter Recommended
All-in One Network	Ethernet 100GbE 2-port QSFP28 Adapter
Storage Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Virtualization Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Host In-Band	Ethernet 10/25GbE 4-Port SFP28 Adapter
BMC OOB Mgt	1G Copper

* Genericized BoM – removing device specifics including dependencies

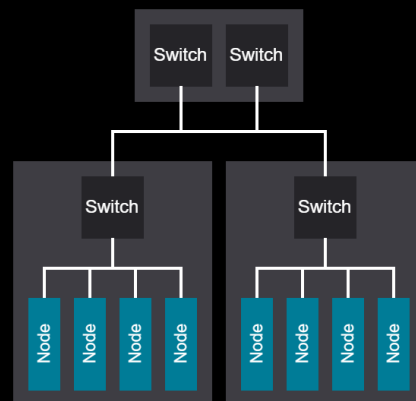
Cluster Network

Basic Network Topologies



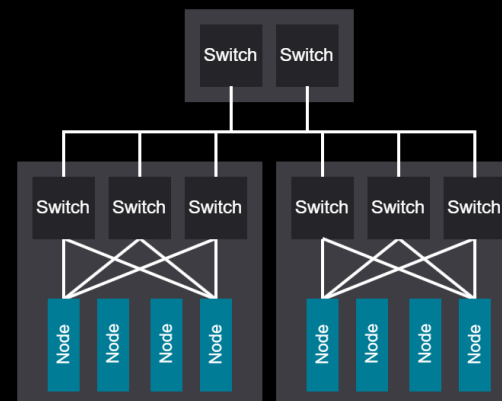
2-Tier Rail Network

- Enables large scalable unit sizes for large jobs or replica sizes
- Efficient for workloads favoring ring-based collectives
- Higher infrastructure costs



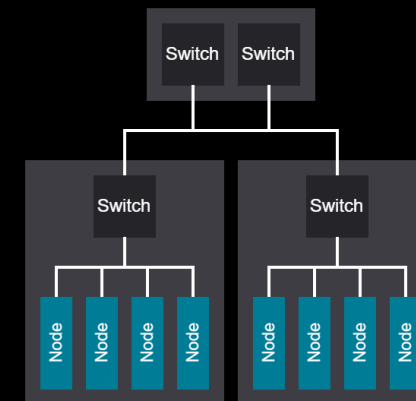
2-Tier Tree Network

- Efficient for small workloads or replicas
- Easy to add capacity with proper planning
- Potentially allows for lower infrastructure costs
- Limited blast radius compared to rail networks



3-Tier Rail TH5/J3 Network

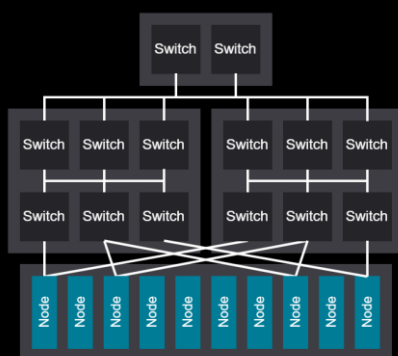
- Spine switches replaced with a 2-tier Jericho3-AI/Ramon3 fabric for increased maximum cluster size
- Deep buffers and scheduled fabric aid greatly with congestion issues in large clusters at a small latency penalty



3-Tier Tree TH5/J3 Network

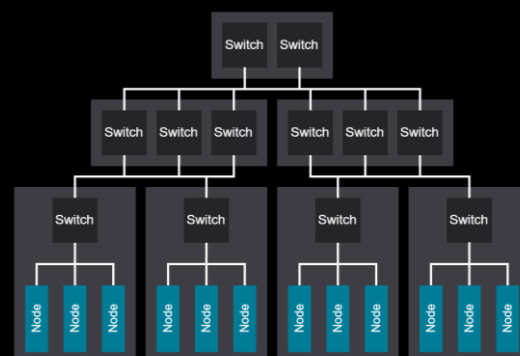
- Same benefits from switch to scheduled spine fabric as with rail, but retains the primary characteristics of tree networks

Basic Network Topologies Continued



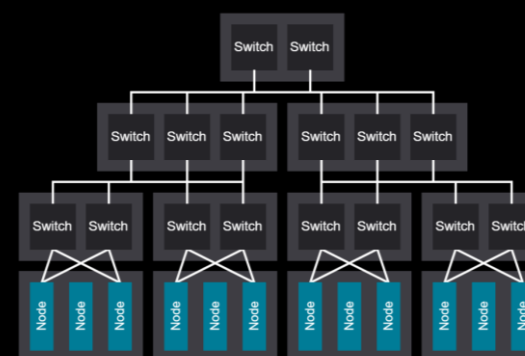
3-Tier Rail Optimized Network

- Allows for massive scalable unit sizes
- Best ring-based collective performance at scale; at the cost of poor any-any performance



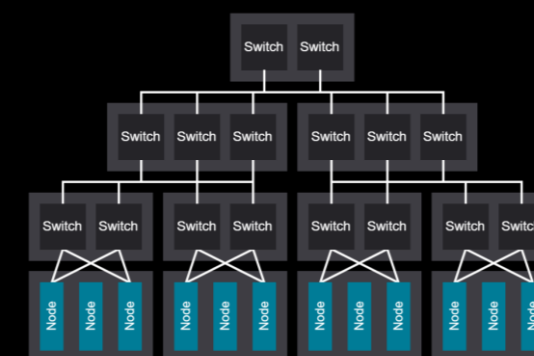
3-Tier Tree Network

- Allows for massive cluster sizes
- Best any-any performance at scale
- Suitable for a “campus style” deployment



3-Tier Hybrid Rail Network

- Allows for massive cluster sizes with large scalable units
- Favors ring-based collectives, but does not sacrifice significant any-any performance on large jobs
- Suitable for a “campus-style” deployment

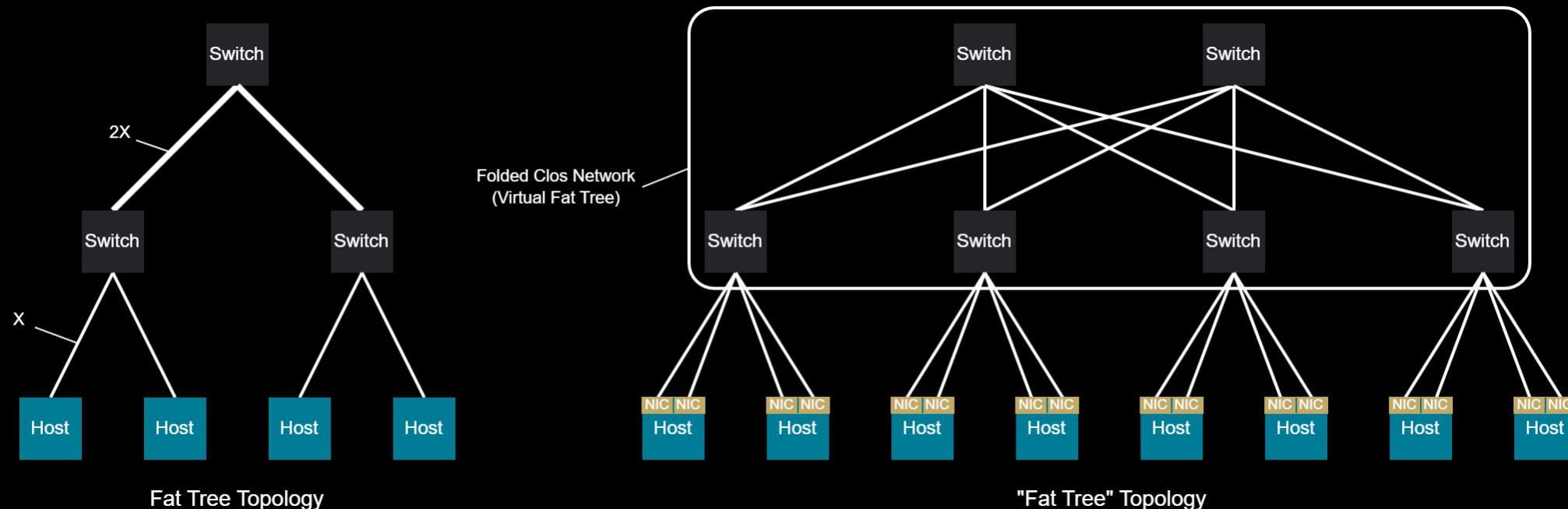


3-Tier Fully Scheduled Rail Network

- Medium sized scalable units
- Excellent congestion performance due to deep buffers and scheduled fabric
- Technical limitations limit cluster size to ~32K GPUs in recommended configuration

Tree and Rail

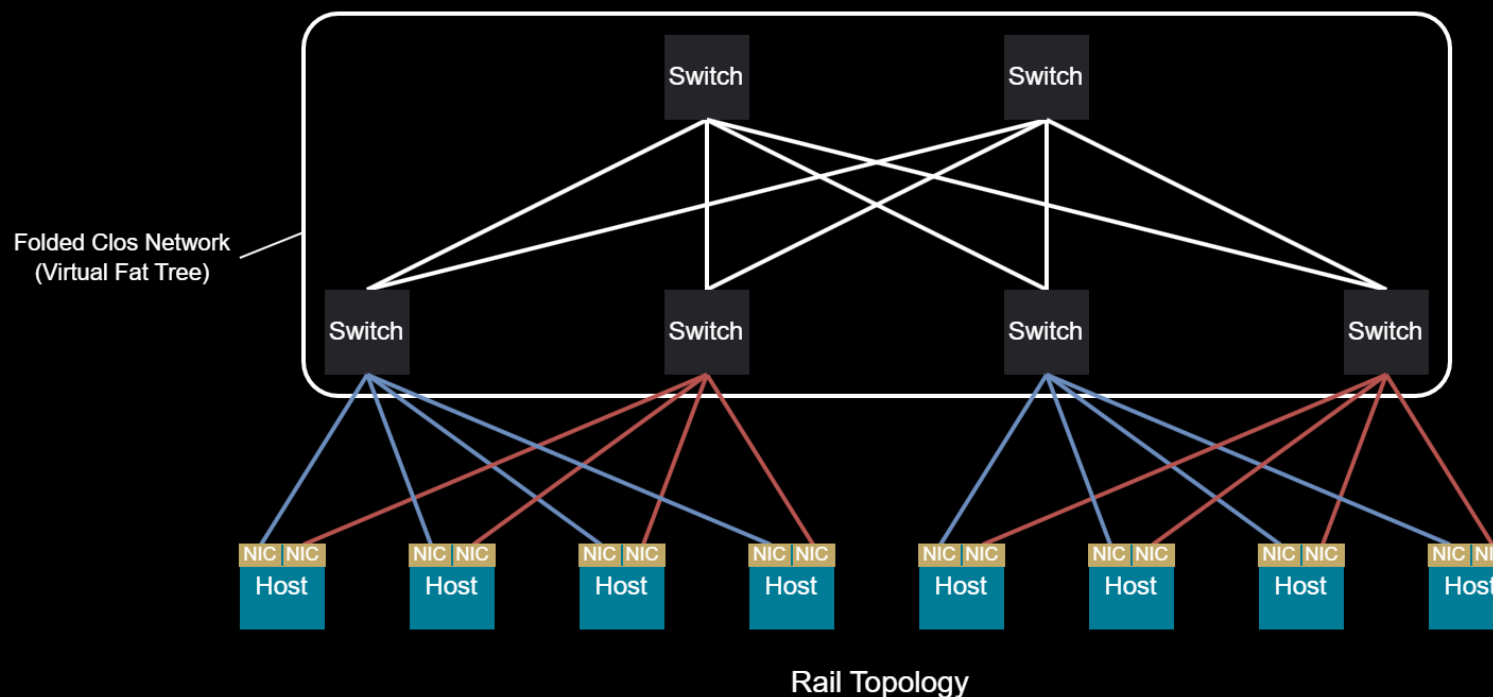
Fat Tree Terminology In Cluster Design



The canonical fat tree topology is a network concept where a switch's connection to upstream peers has at least parity bandwidth with the total aggregate bandwidth of its downstream connections. This causes links between switches to become “fatter” as they get closer to the core.

The “fat tree” topology for AI/ML clusters instead refers to how a host is connected to its upstream switches; in this case all host NICs terminate on the same switch. It can also be considered 1-rail network. The network itself is generally a 3-stage or 5-stage folded Clos network due the fixed radix of network switches.

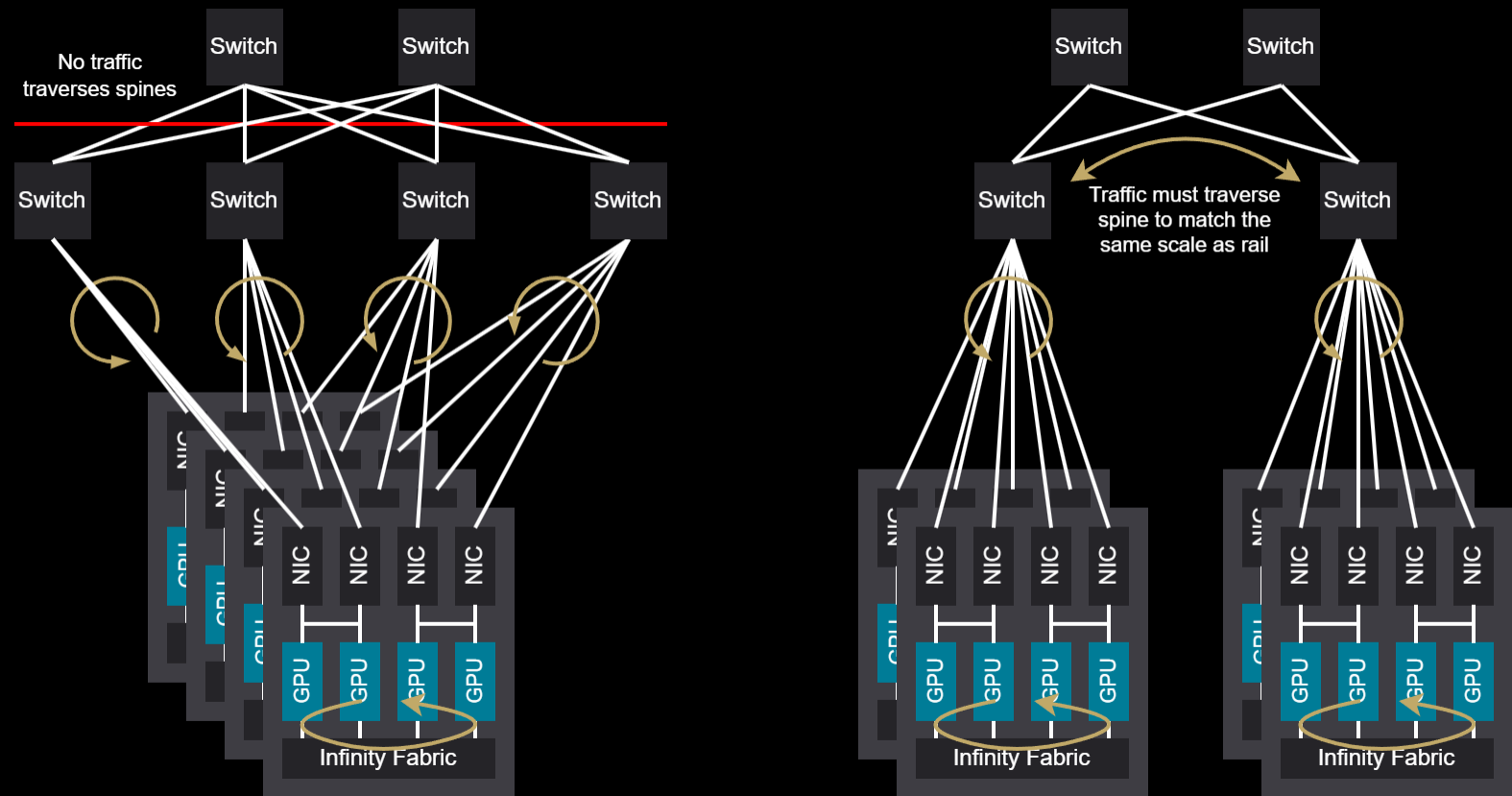
Rail Networks In Cluster Design



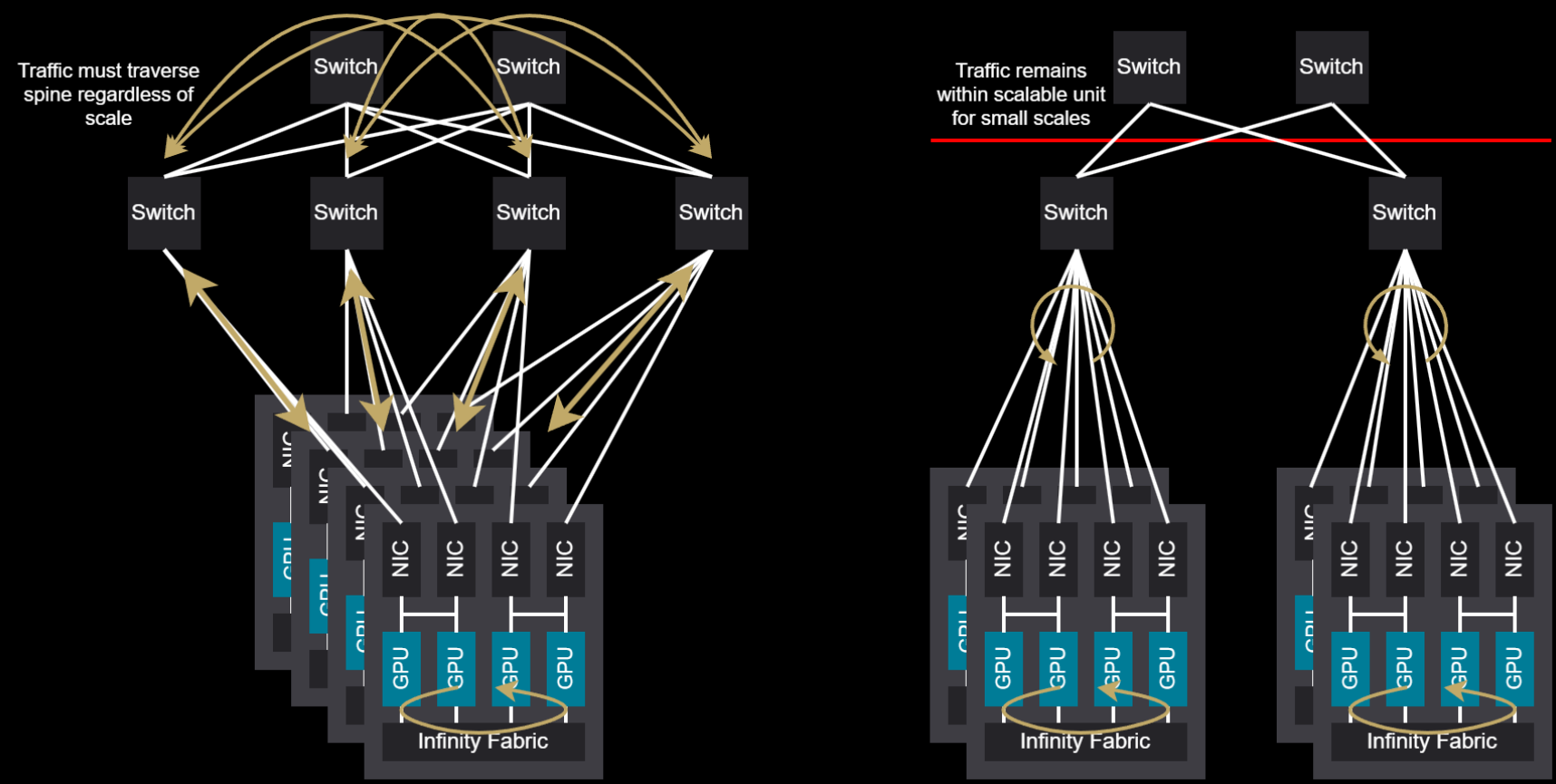
Rail networks leverage the same folded Clos network as tree networks, but host connections are instead aggregated onto switches based on NIC rank. These shared ranks are referred to as rails and allow the network to provide preferential latency for connections which share the same rail. The downside to this design is any traffic which needs to cross rails/ranks must traverse either the network spine layer, or Infinity Fabric (PXN).

The above example shows an example 2-rail network, with the rails colored blue and red to differentiate them.

Rail Enables Larger Single Hop Ring Domains

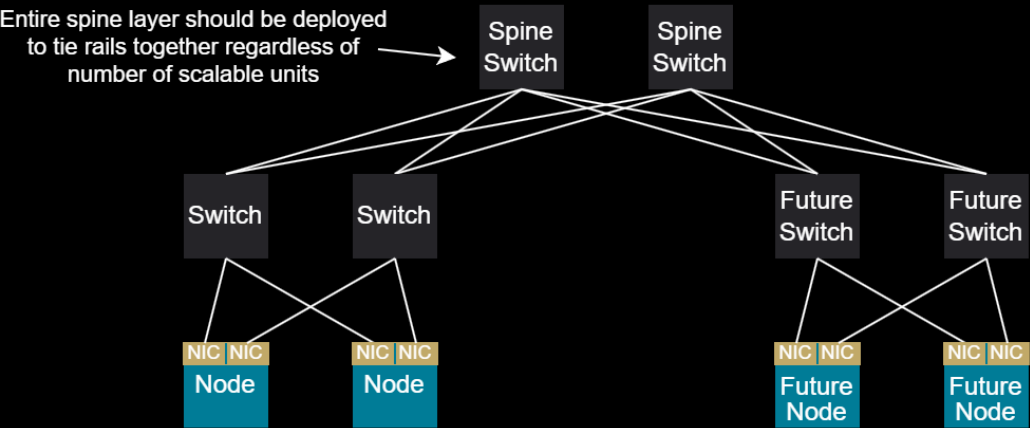
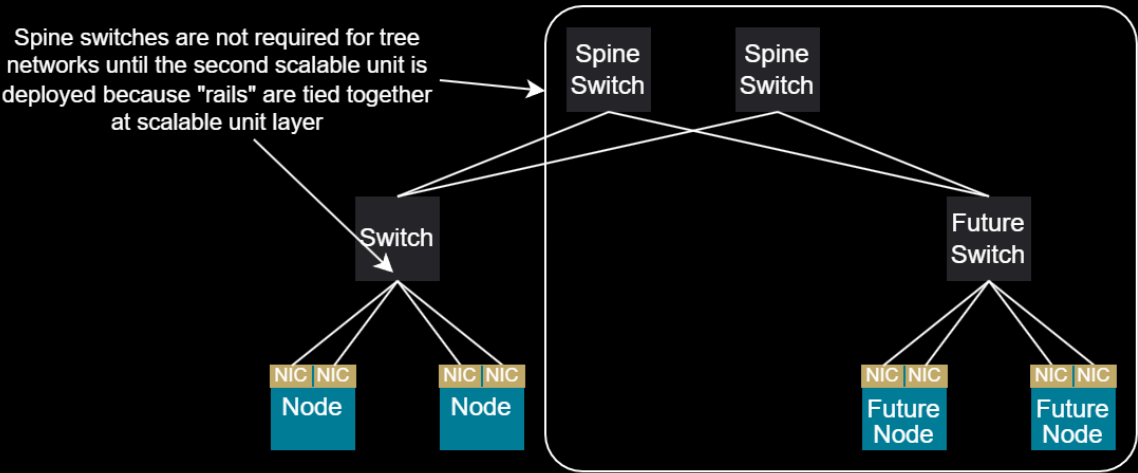


Tree Handles Cross-Rank Traffic Better

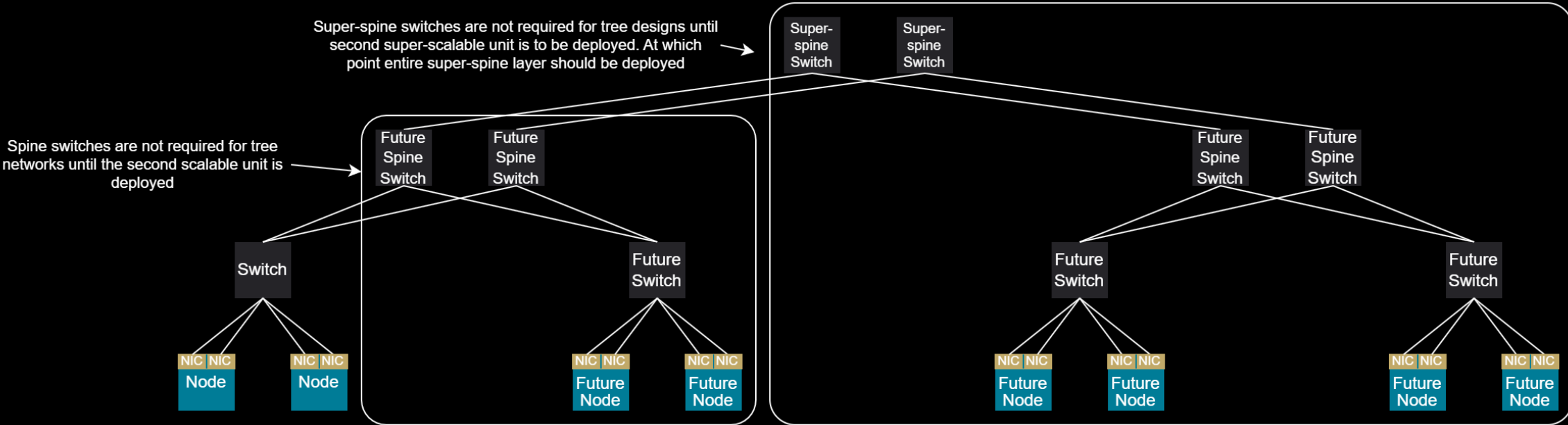


Scaling Networks

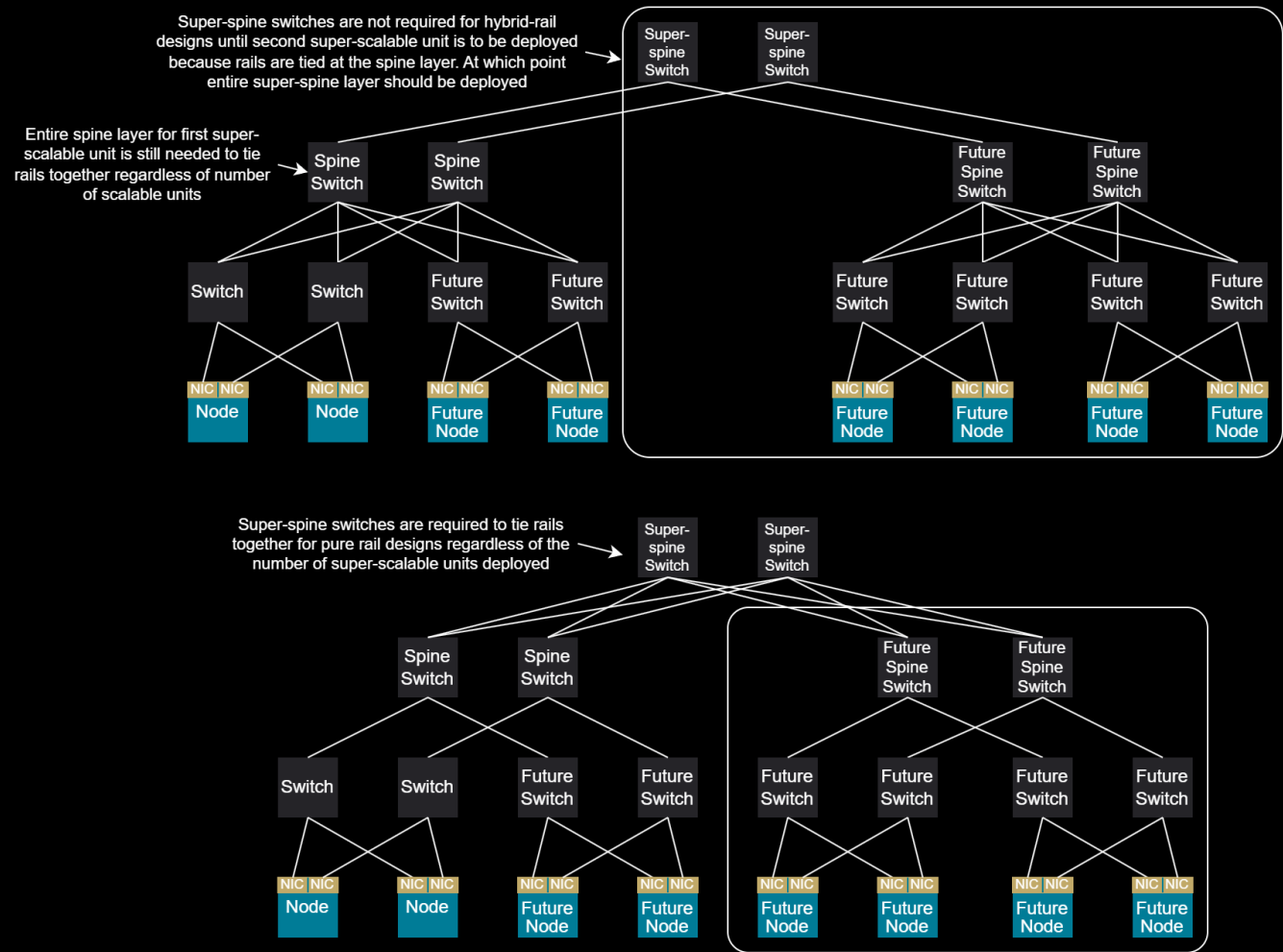
Cluster Backend Deployment Strategy for 2-tier Networks



Cluster Backend Deployment Strategy for 3-tier Tree Networks



Cluster Backend Deployment Strategy for 3-tier Rail Networks



Network Subscription

Subscription is the relationship between what is provided by the upstream network and what is required by the downstream network in demand side.

It is typically represented as a ratio:

$$\text{Downstream Demand} : \text{Upstream Capacity}$$

1:1 subscribed network would be equal downstream capacity to upstream capacity.

Example: 1:1.16 = (1 downstream demand : 1.16 upstream capacity. In this example there is .16 more upstream capacity)

Or as a percentage:

$$\text{Subscription Rate} = \frac{\text{Downstream Demand}}{\text{Upstream Capacity}}$$

80% subscription ratio could be referred to as “20% undersubscribed”, or a 120% subscription ratio could be referred to as “20% oversubscribed”.

Network Design Examples

Network Topology Design Notations:

Design note:

Designs included are based on scheduled fabrics switch type (Arista, Ciena, Nokia) or 51.2T switch type (Arista, Cisco, Dell, Juniper)

Vendors/switch models vary for port count and features – please consult desired vendor port count directly to confirm.

Scalable Units/PODs note:

Diagrams presented are designed around a Scalable Unit or POD – which can determine overall network end to end latency and AI use cases.

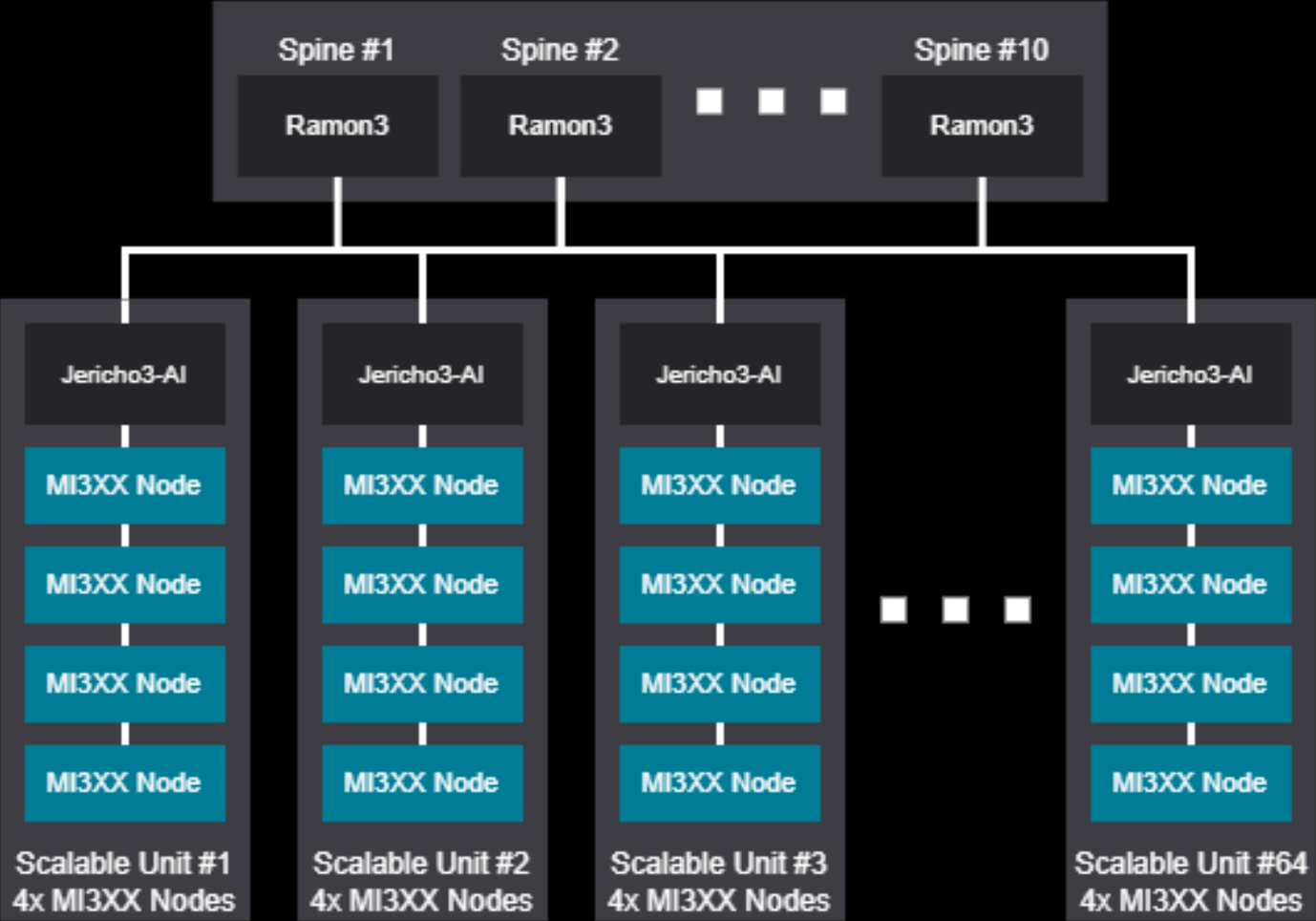
Certain ML/AI workloads may require change of scalable unit size. Please consult with AMD Architecture as required.

2K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 2048 GPU (256 Nodes)

Tree Design

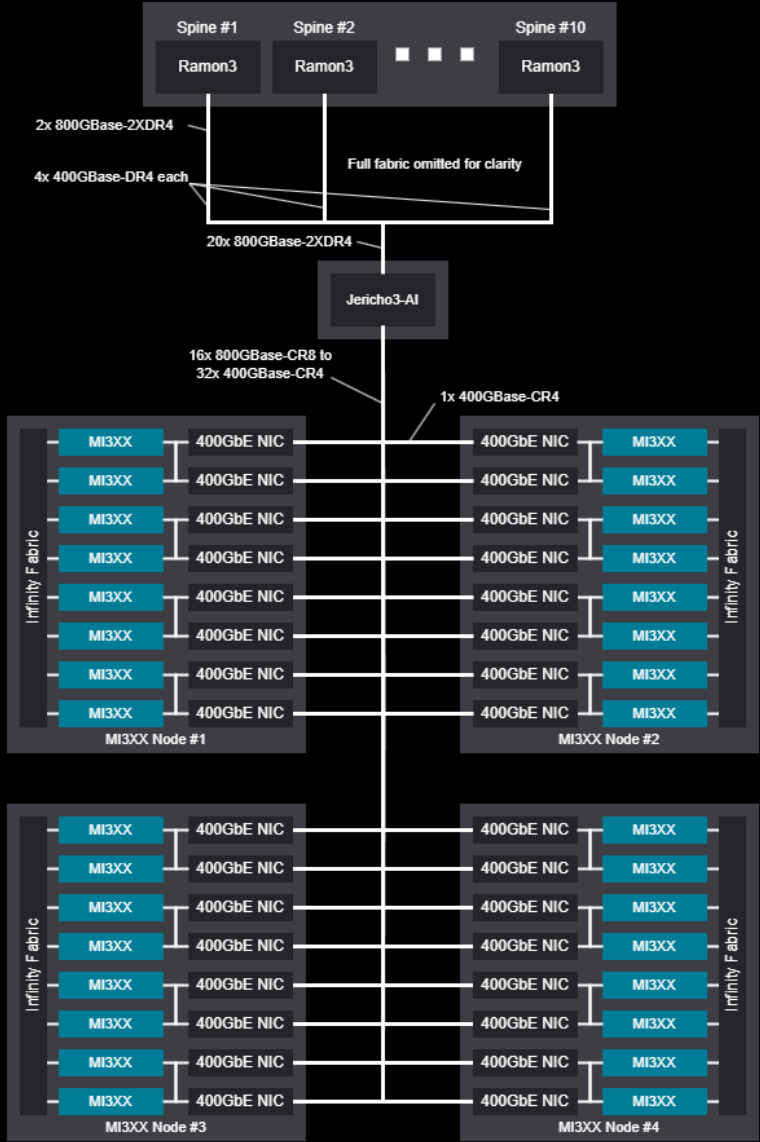
Jericho/Ramon



Network Diagram - 2048 GPU (256 Nodes)

Tree Scalable Unit

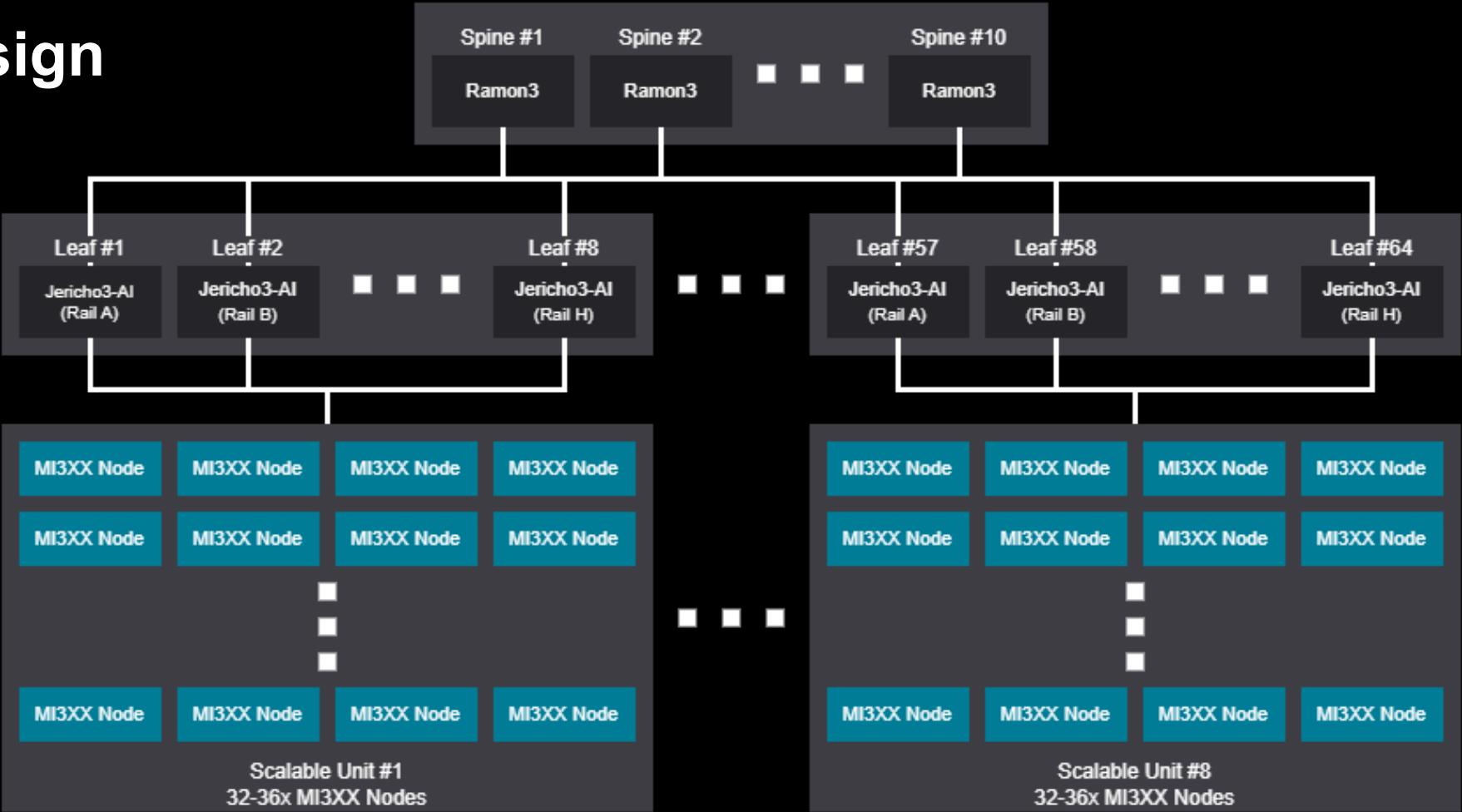
Jericho/Ramon



Network Diagram - 2048-2304 GPU (256-288 Nodes)

Rail Design

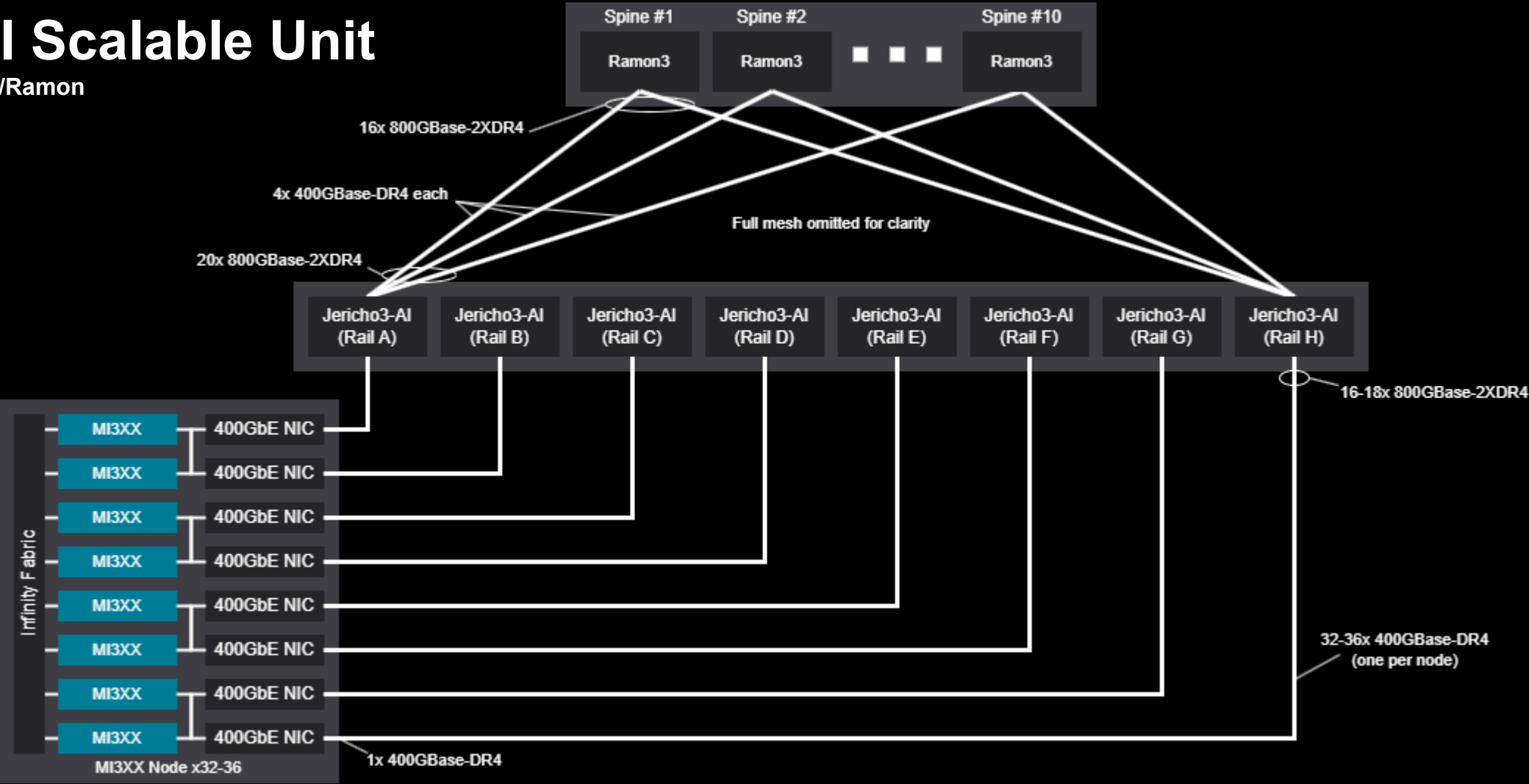
Jericho/Ramon



Network Diagram - 2048-2304 GPU (256-288 Nodes)

Rail Scalable Unit

Jericho/Ramon



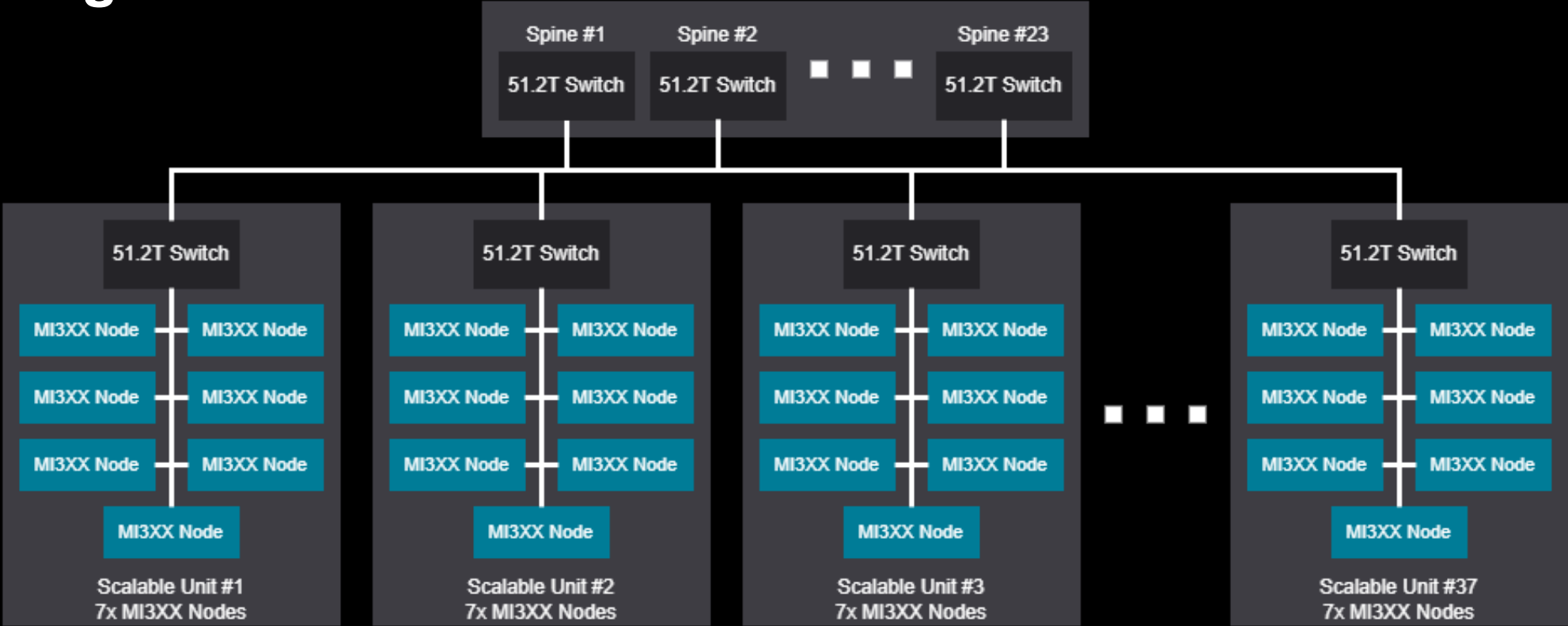
2K GPU Topology Design Examples

51.2T

Network Diagram - 2072 GPU (259 Nodes)

Tree Design

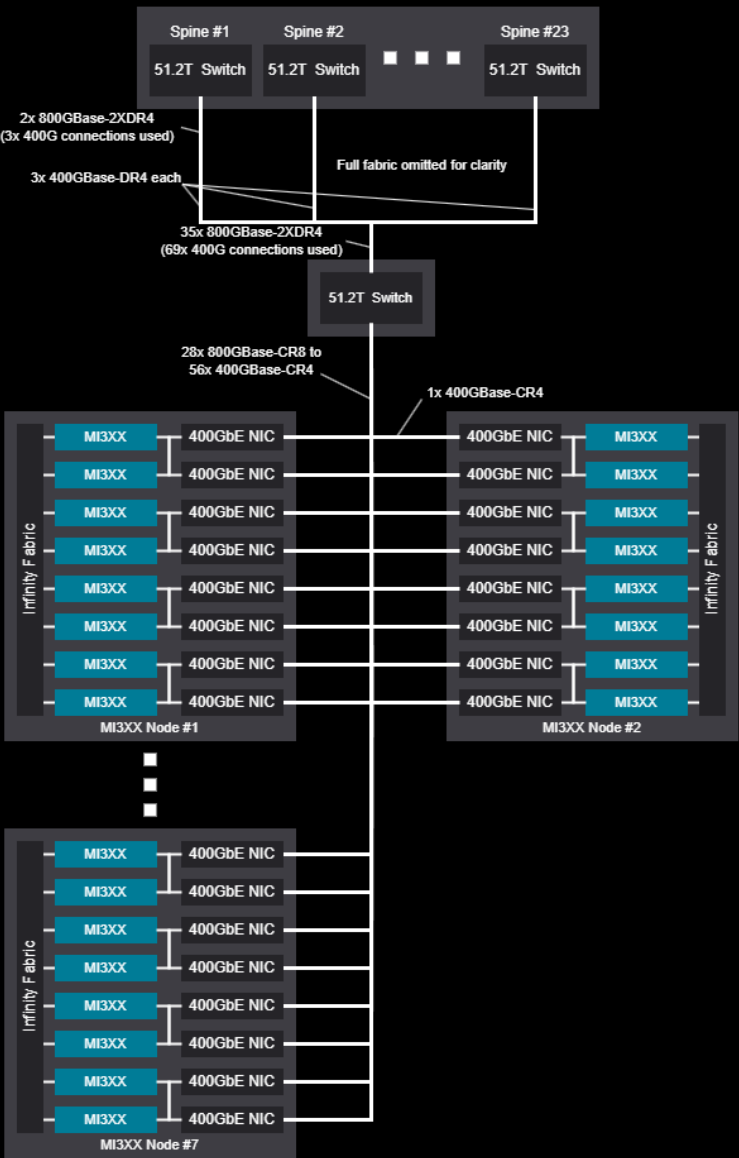
51.2T



Network Diagram - 2072 GPU (259 Nodes)

Tree Scalable Unit

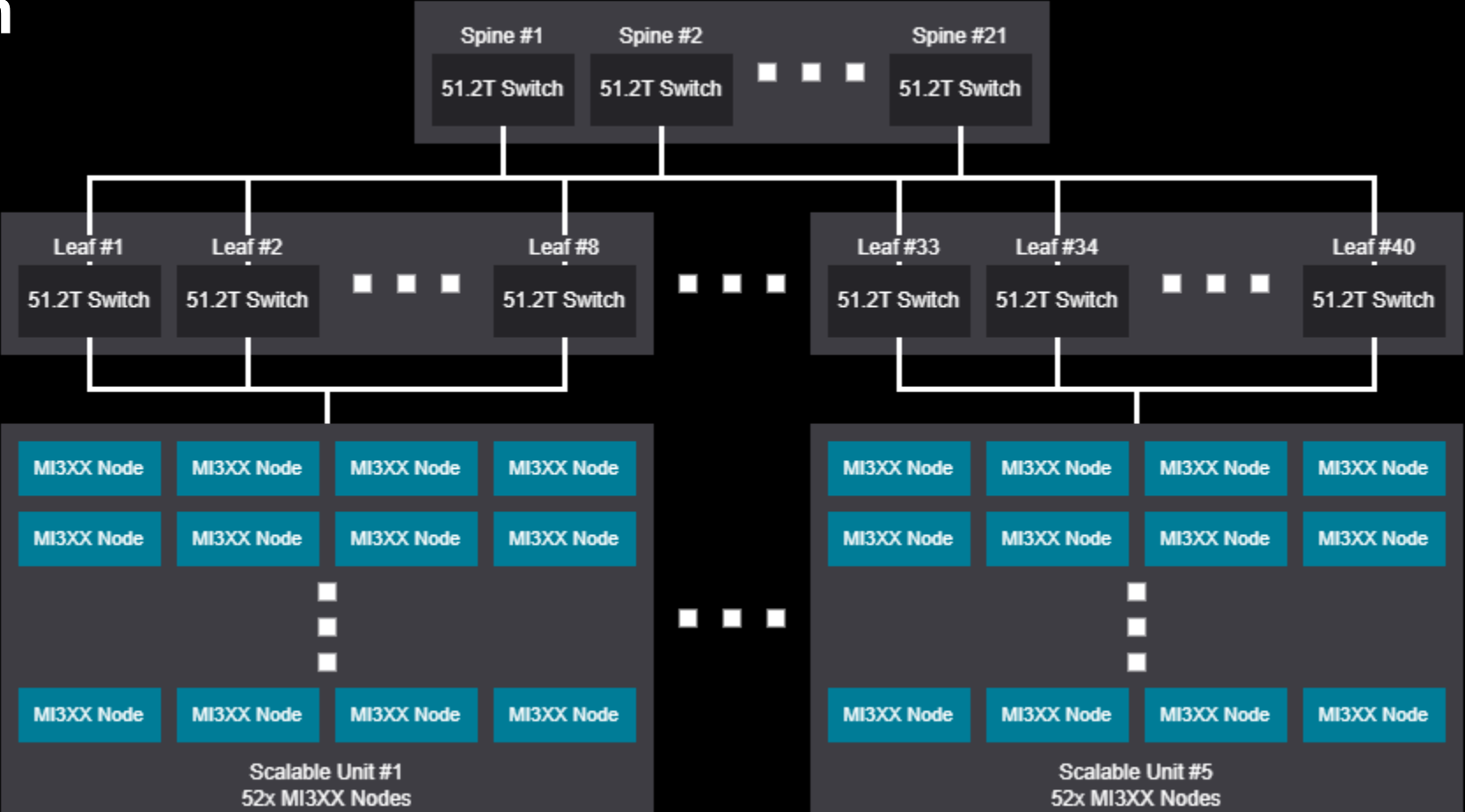
51.2T



Network Diagram - 2080 GPU (260 Nodes)

Rail Design

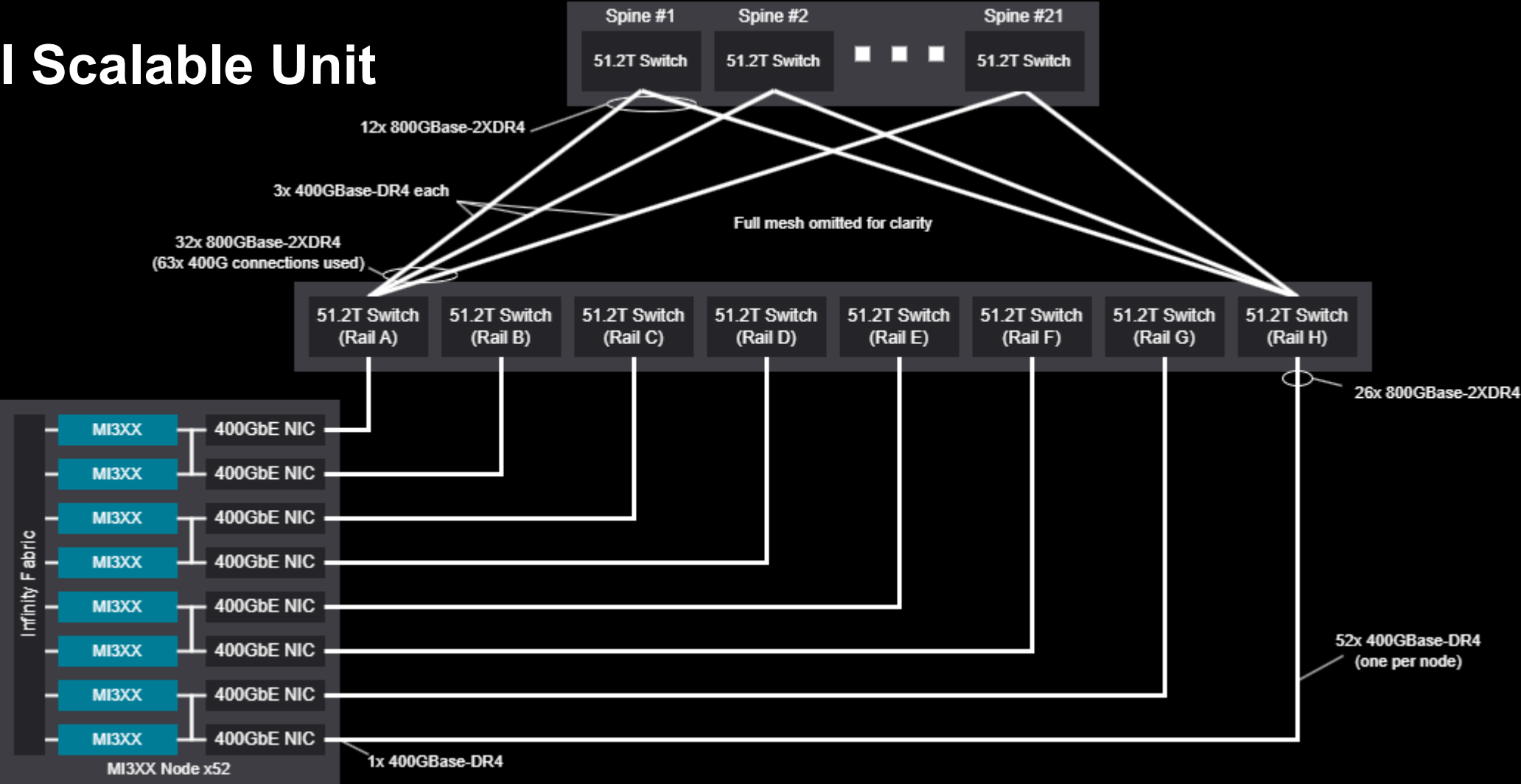
51.2T



Network Diagram - 2080 GPU (260 Nodes)

Rail Scalable Unit

51.2T

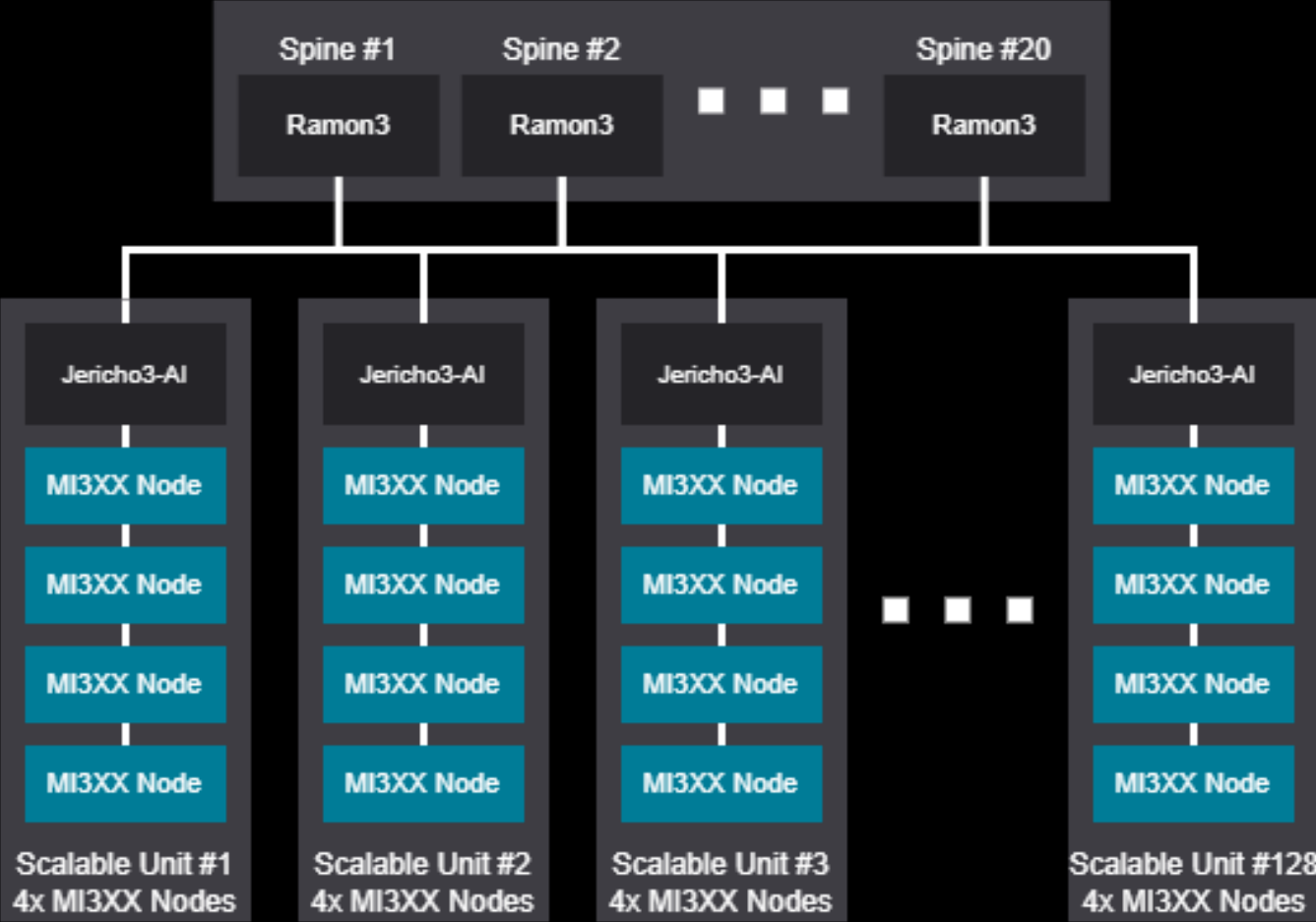


4K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 4096 GPU (512 Nodes)

Tree Design

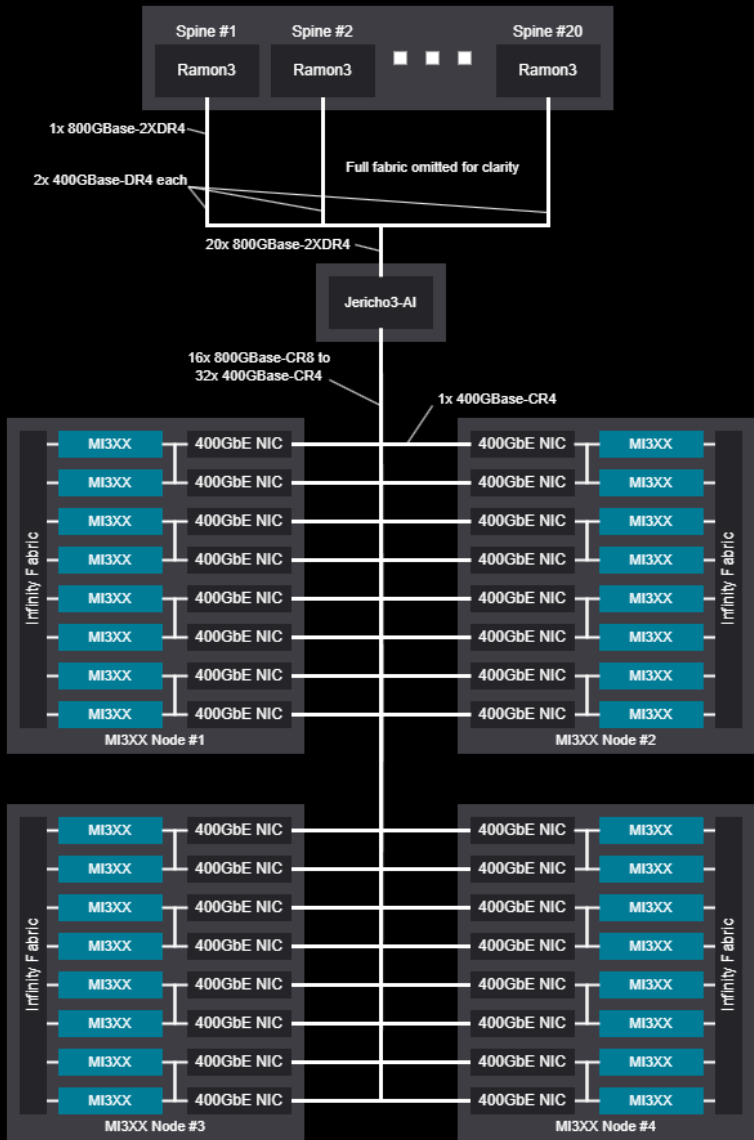
Jericho/Ramon



Network Diagram - 4096 GPU (512 Nodes)

Tree Scalable Unit

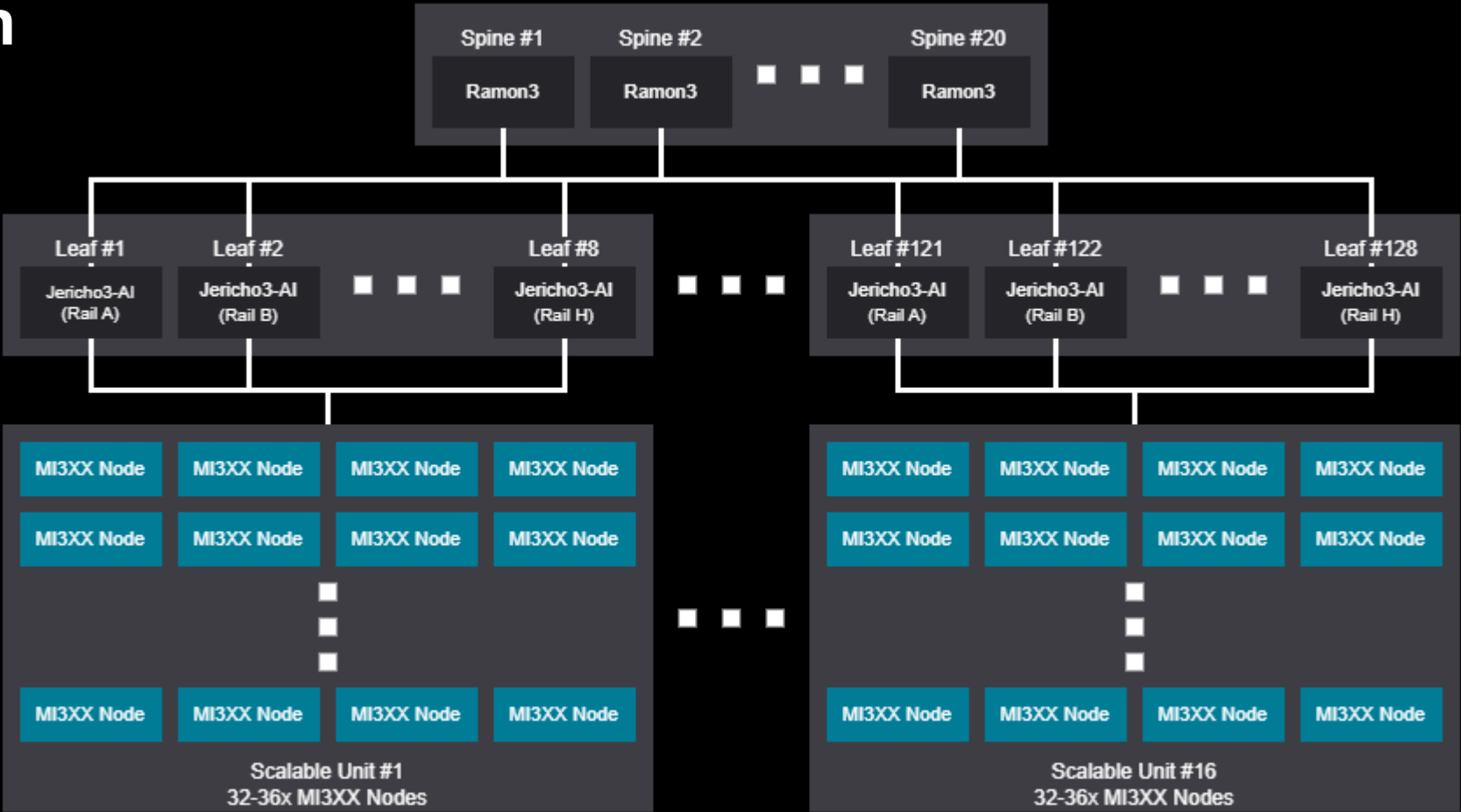
Jericho/Ramon



Network Diagram - 4096-4608 GPU (512-576 Nodes)

Rail Design

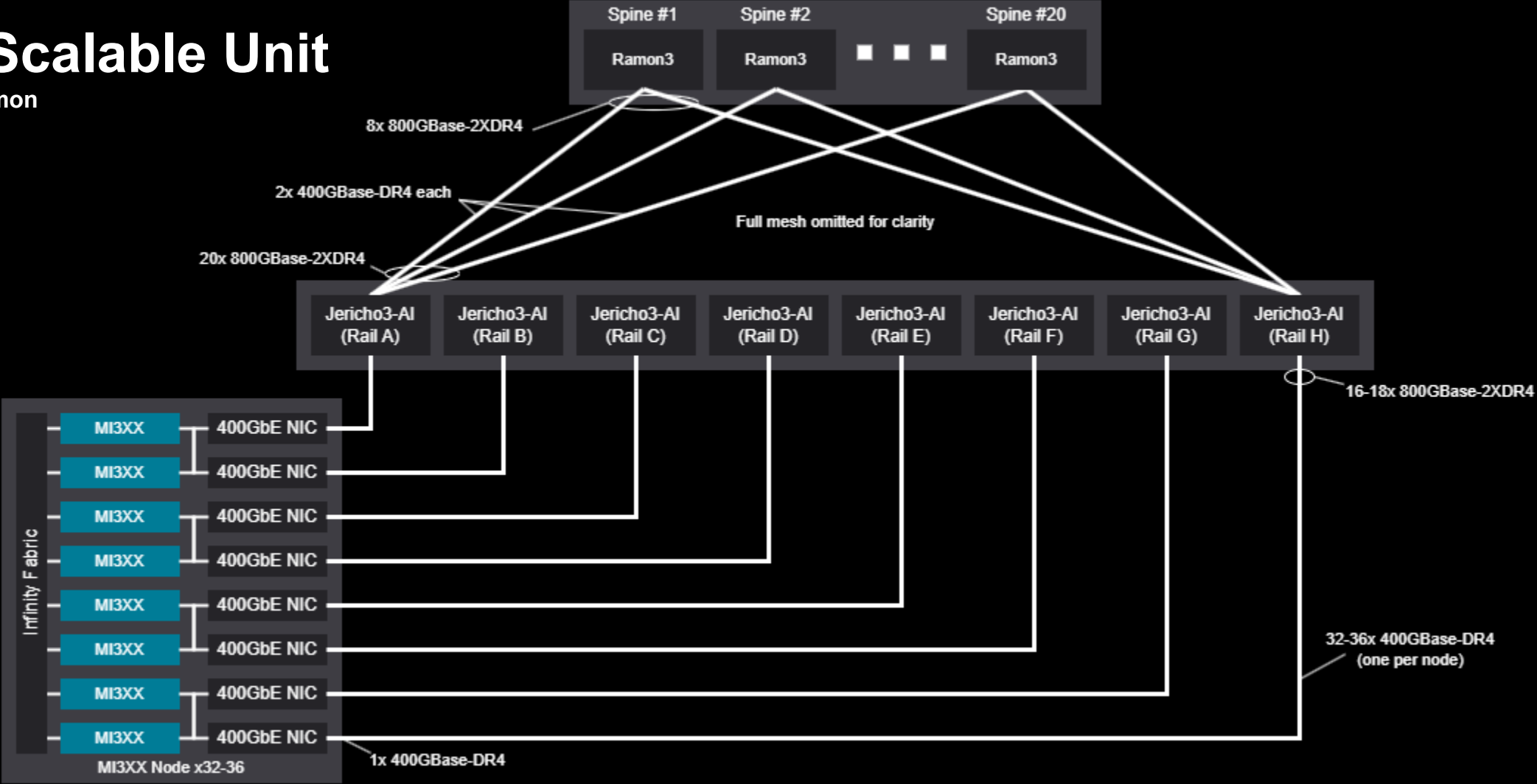
Jericho/Ramon



Network Diagram - 4096-4608 GPU (512-576 Nodes)

Rail Scalable Unit

Jericho/Ramon



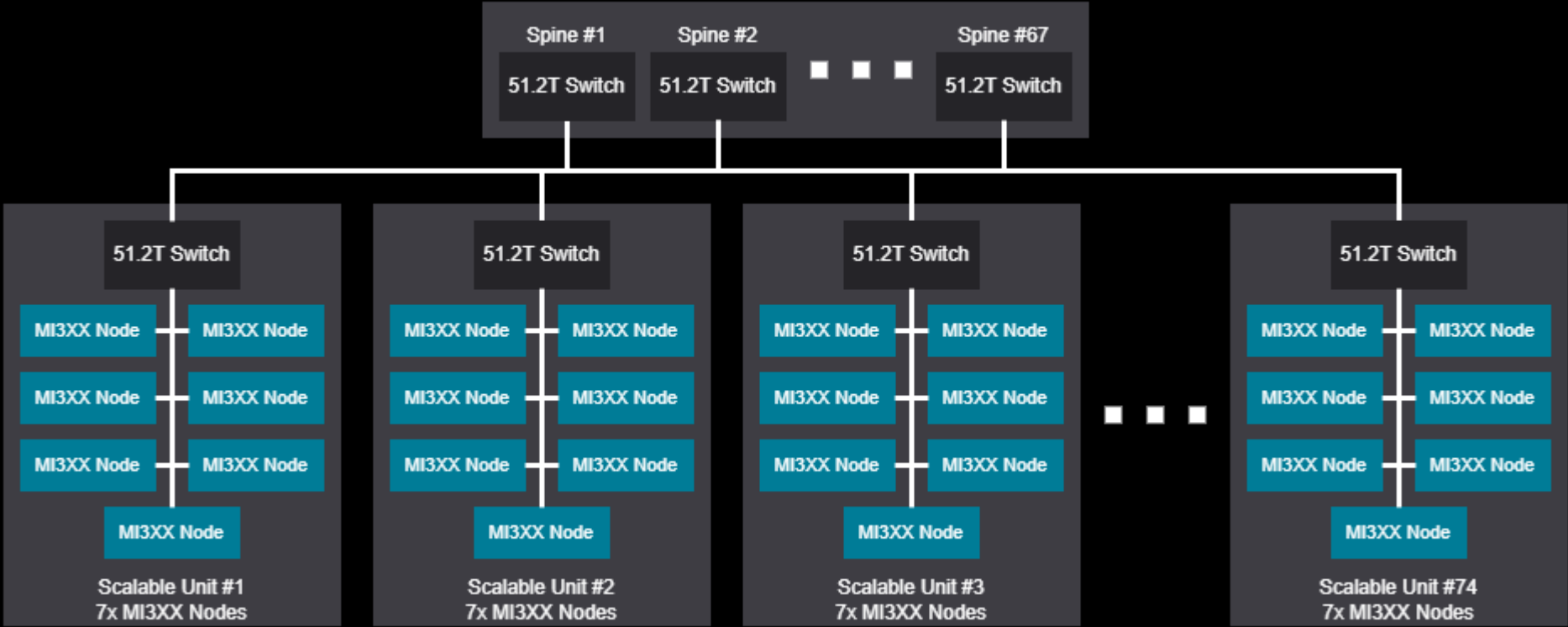
4K GPU Topology Design Examples

51.2T

Network Diagram - 4144 GPU (518 Nodes)

Tree Design

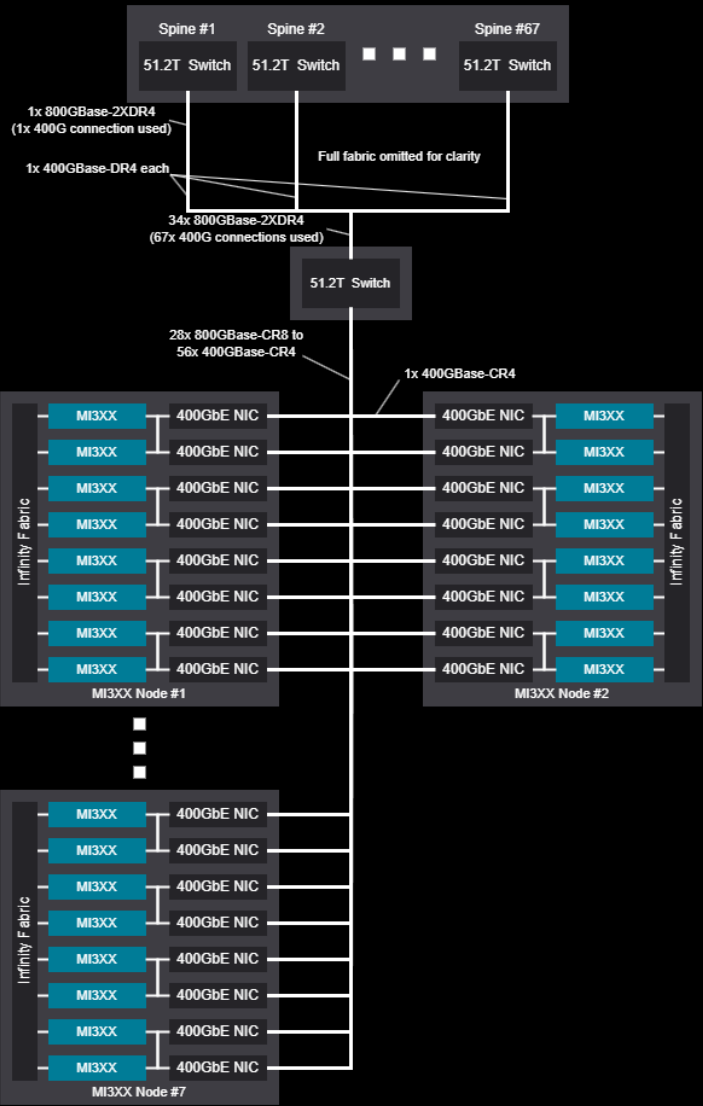
51.2T



Network Diagram - 4144 GPU (518 Nodes)

Tree Scalable Unit

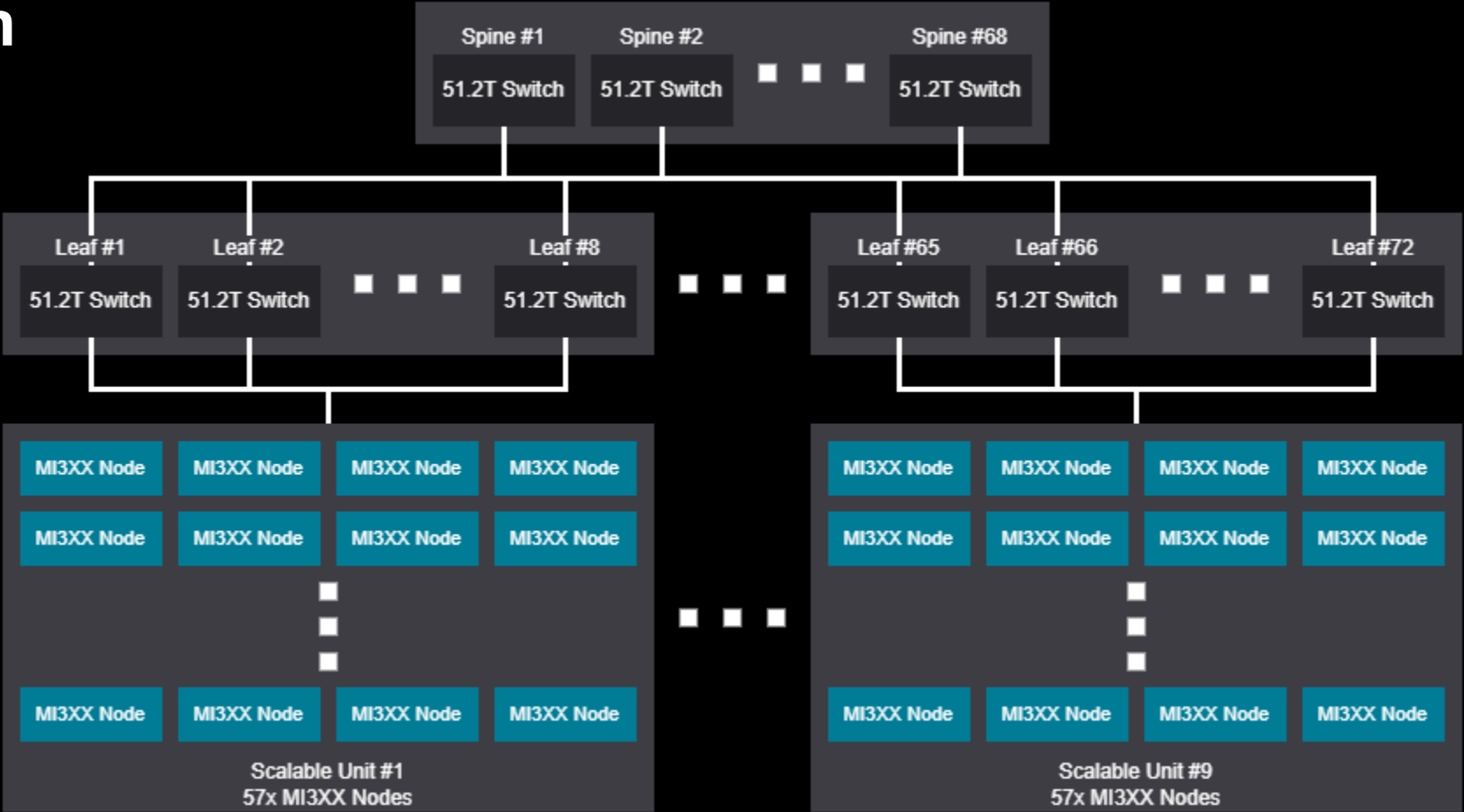
51.2T



Network Diagram - 4104 GPU (513 Nodes)

Rail Design

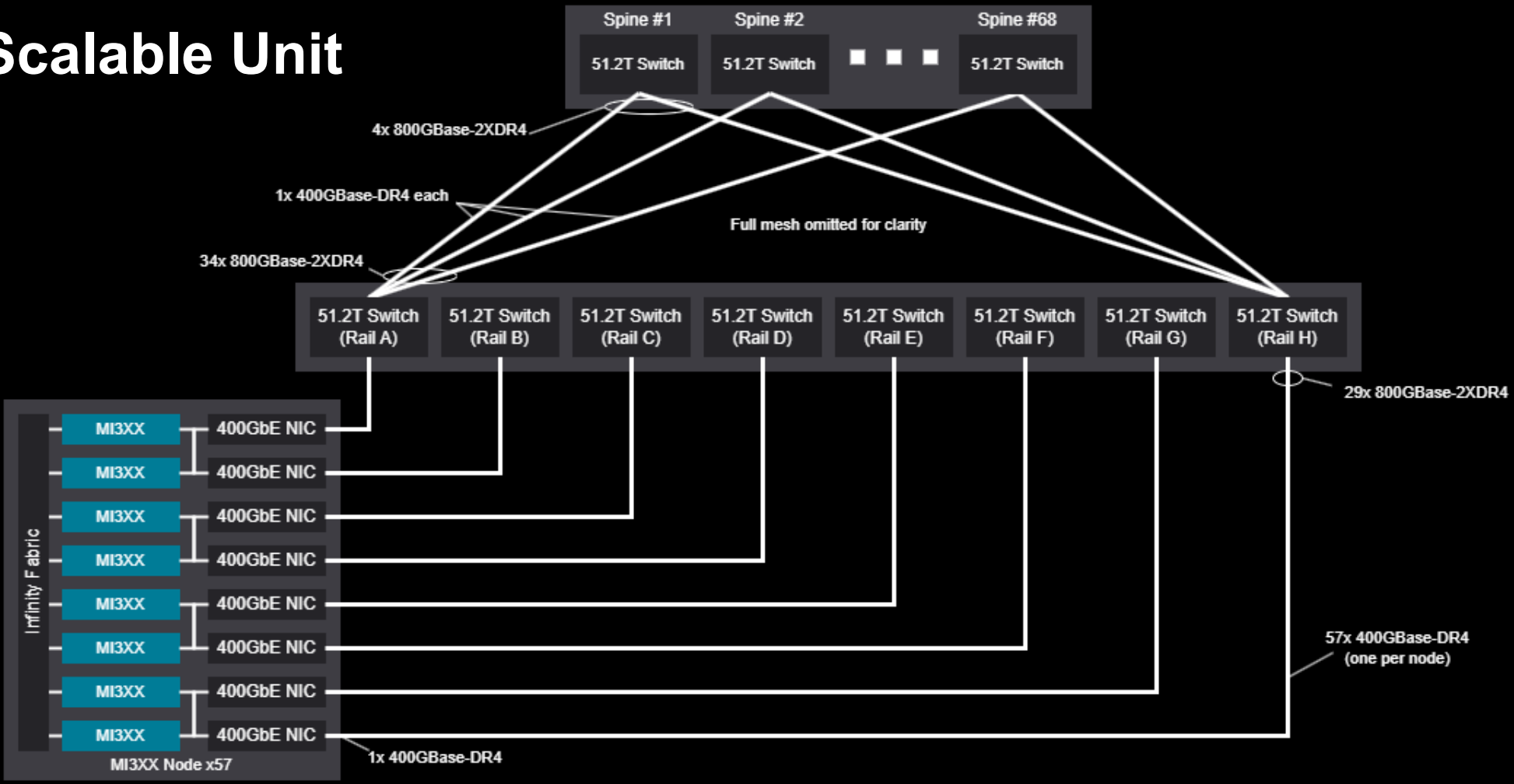
51.2T



Network Diagram - 4104 GPU (513 Nodes)

Rail Scalable Unit

51.2T

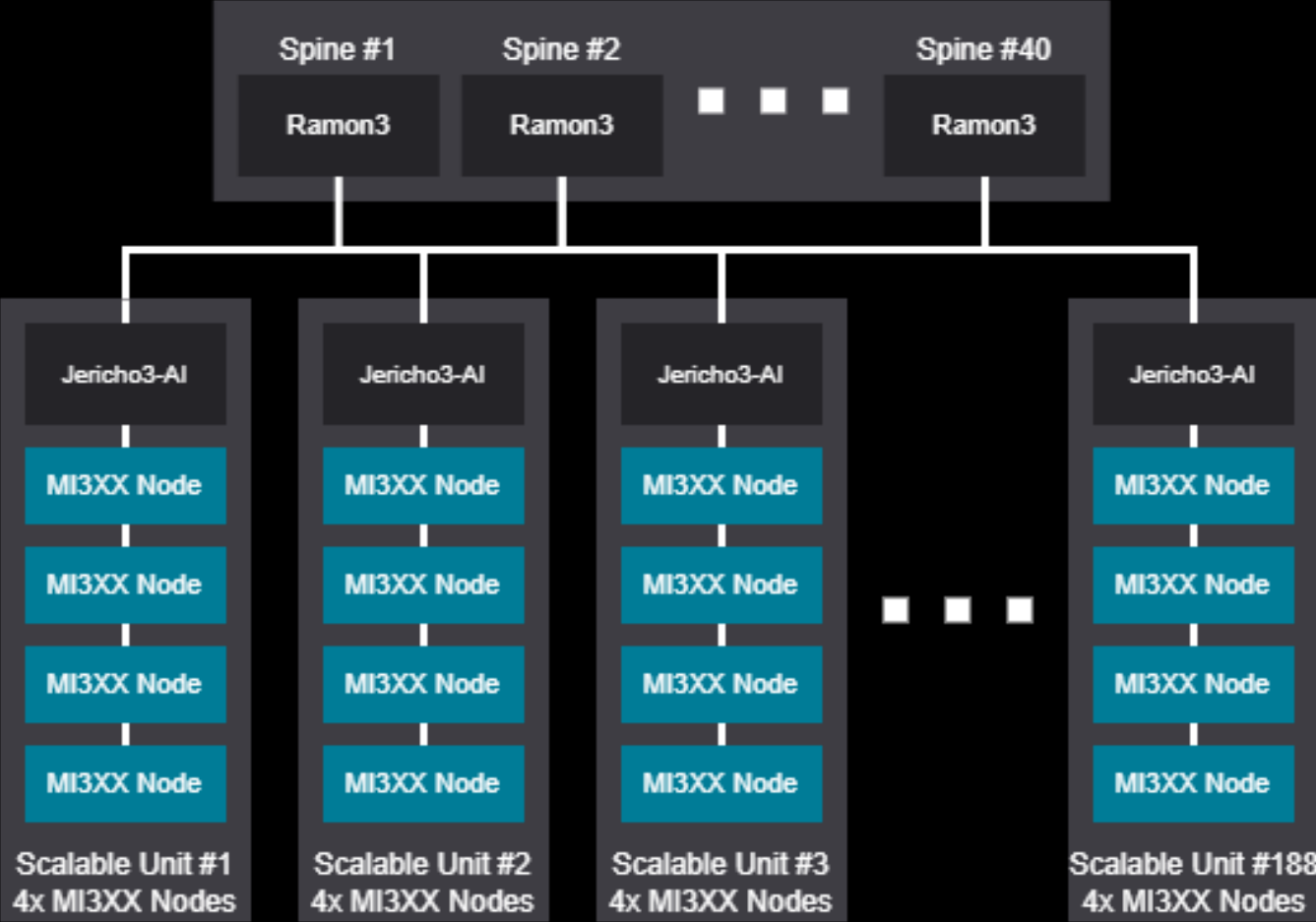


6K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 6016 GPU (752 Nodes)

Tree Design

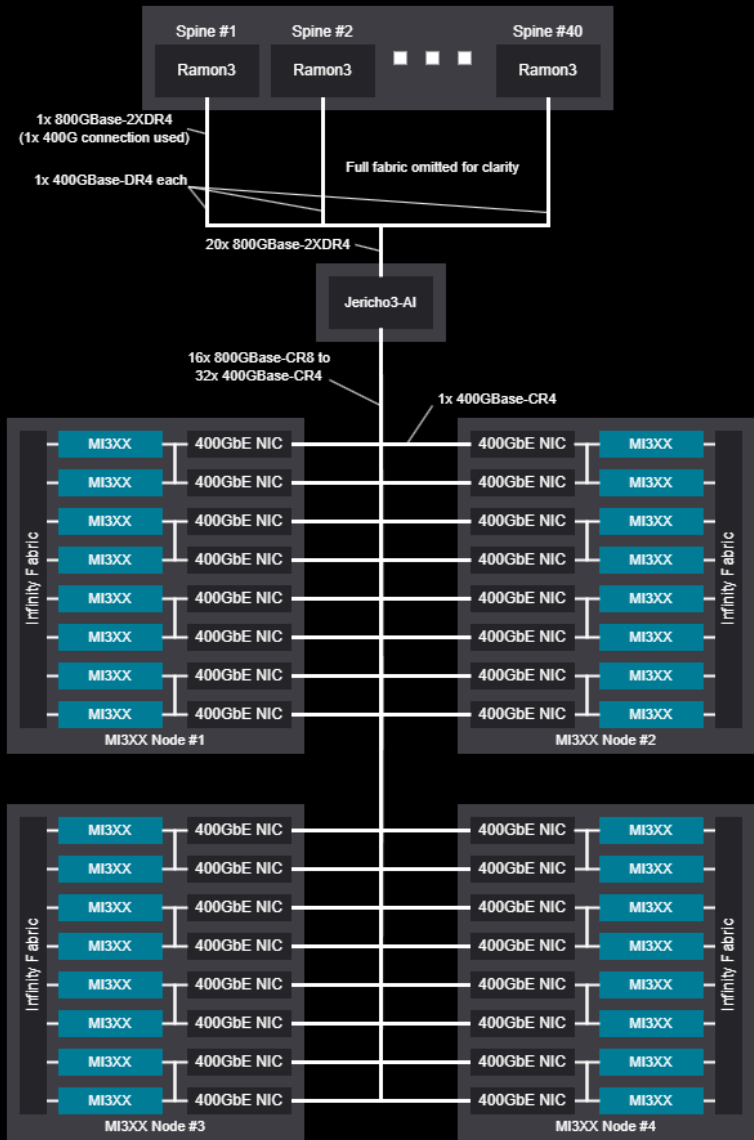
Jericho/Ramon



Network Diagram - 6016 GPU (752 Nodes)

Tree Scalable Unit

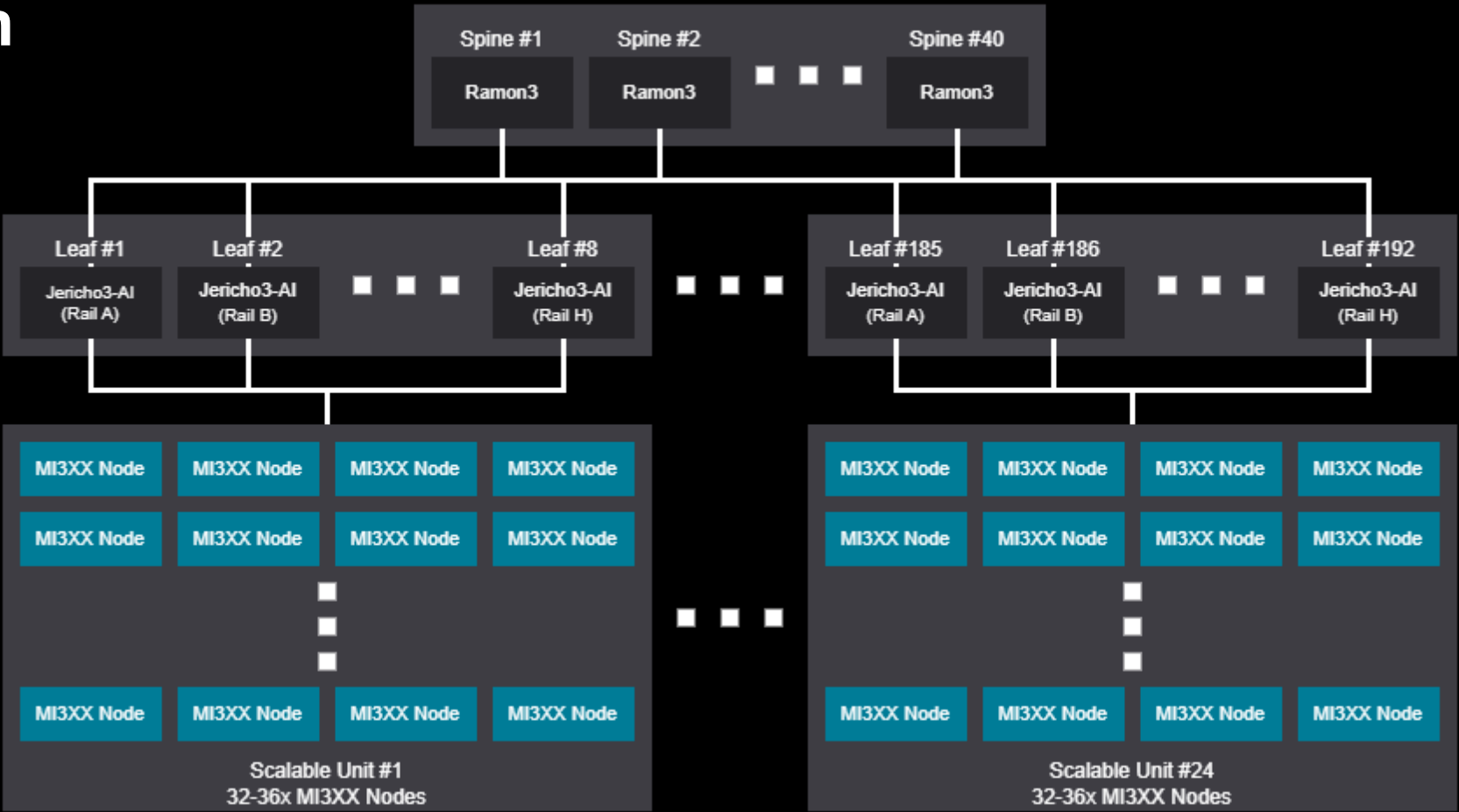
Jericho/Ramon



Network Diagram - 6144-6912 GPU (768-864 Nodes)

Rail Design

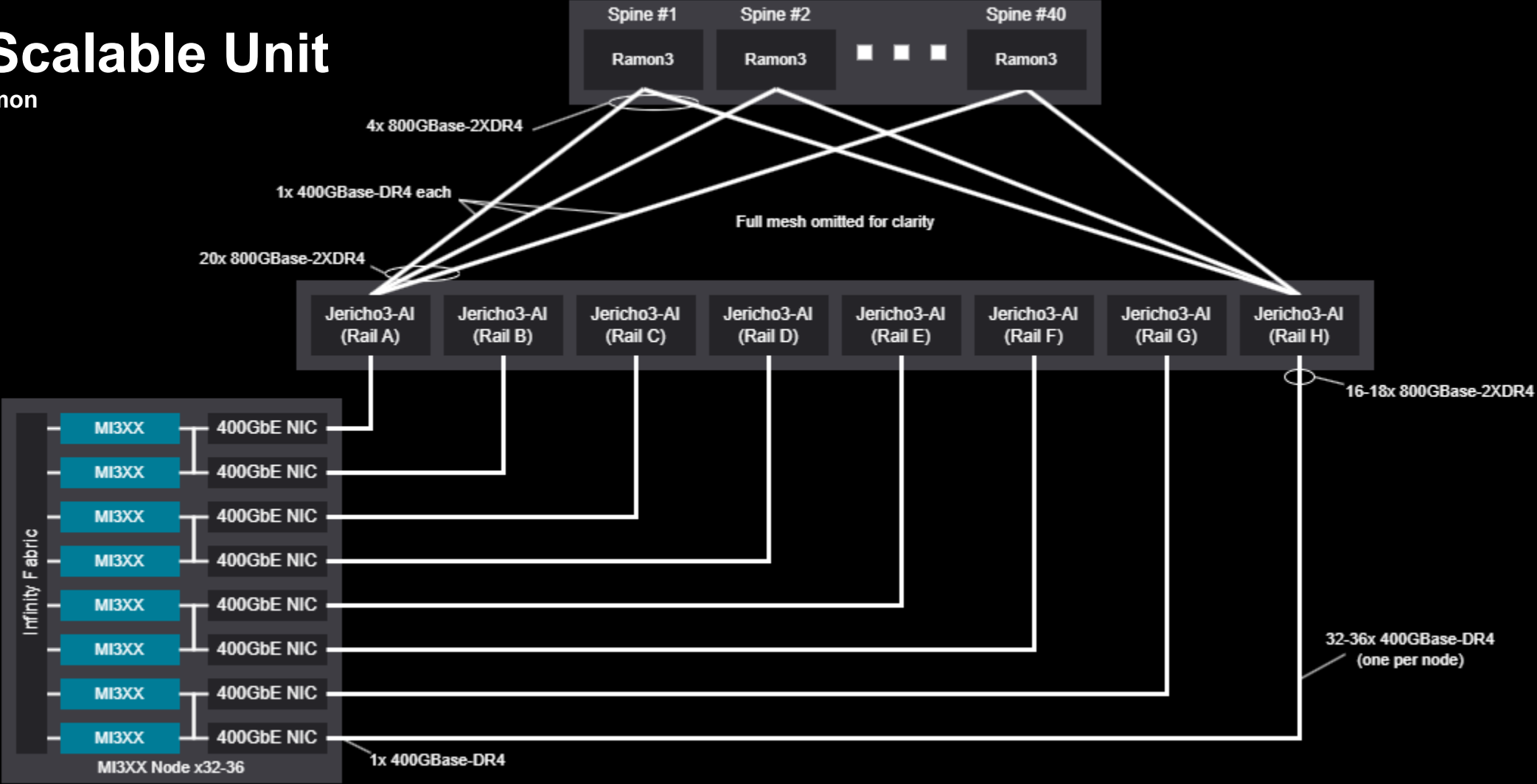
Jericho/Ramon



Network Diagram - 6144-6912 GPU (768-864 Nodes)

Rail Scalable Unit

Jericho/Ramon



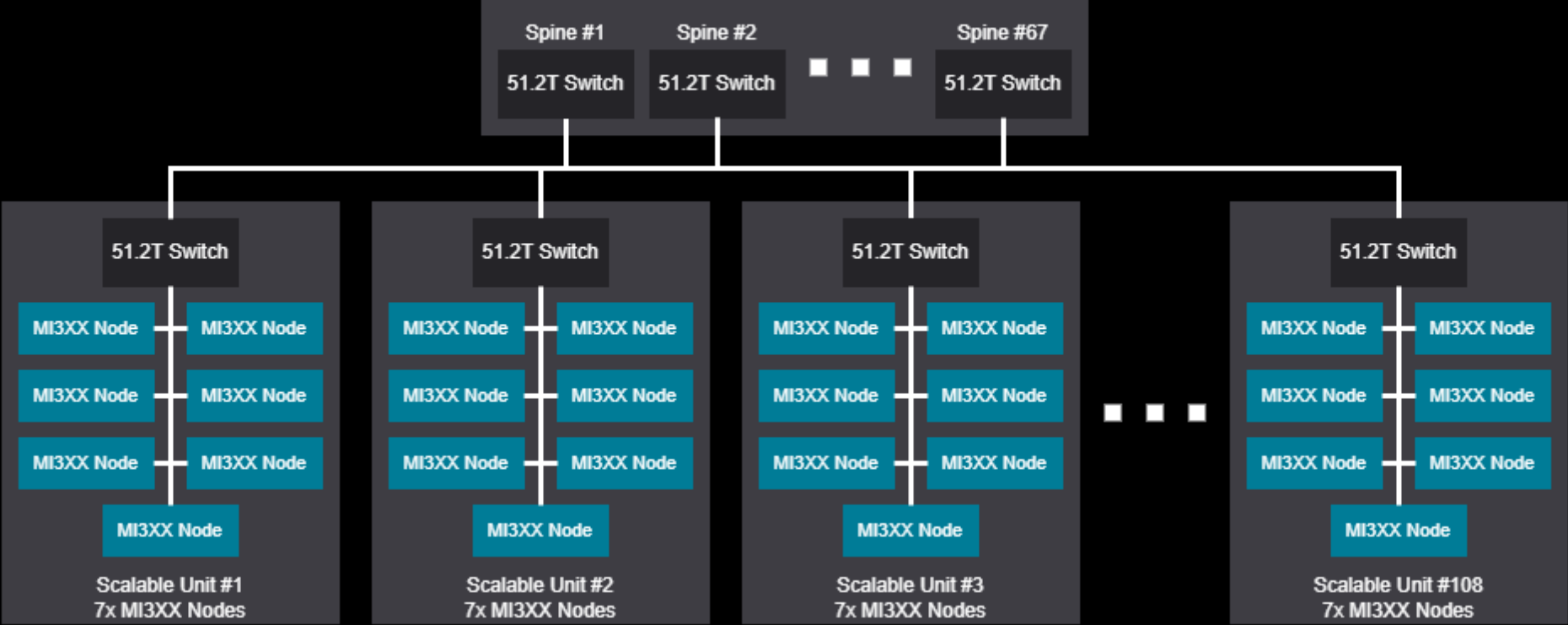
6K GPU Topology Design Examples

51.2T

Network Diagram - 6048 GPU (756 Nodes)

Tree Design

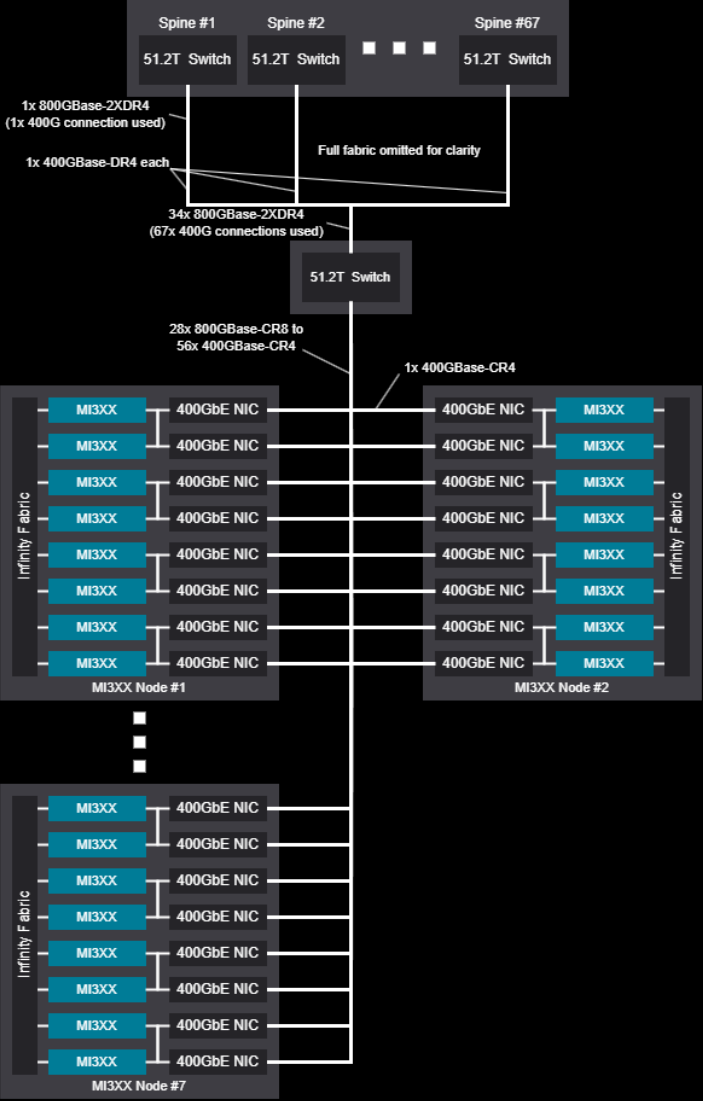
51.2T



Network Diagram - 6048 GPU (756 Nodes)

Tree Scalable Unit

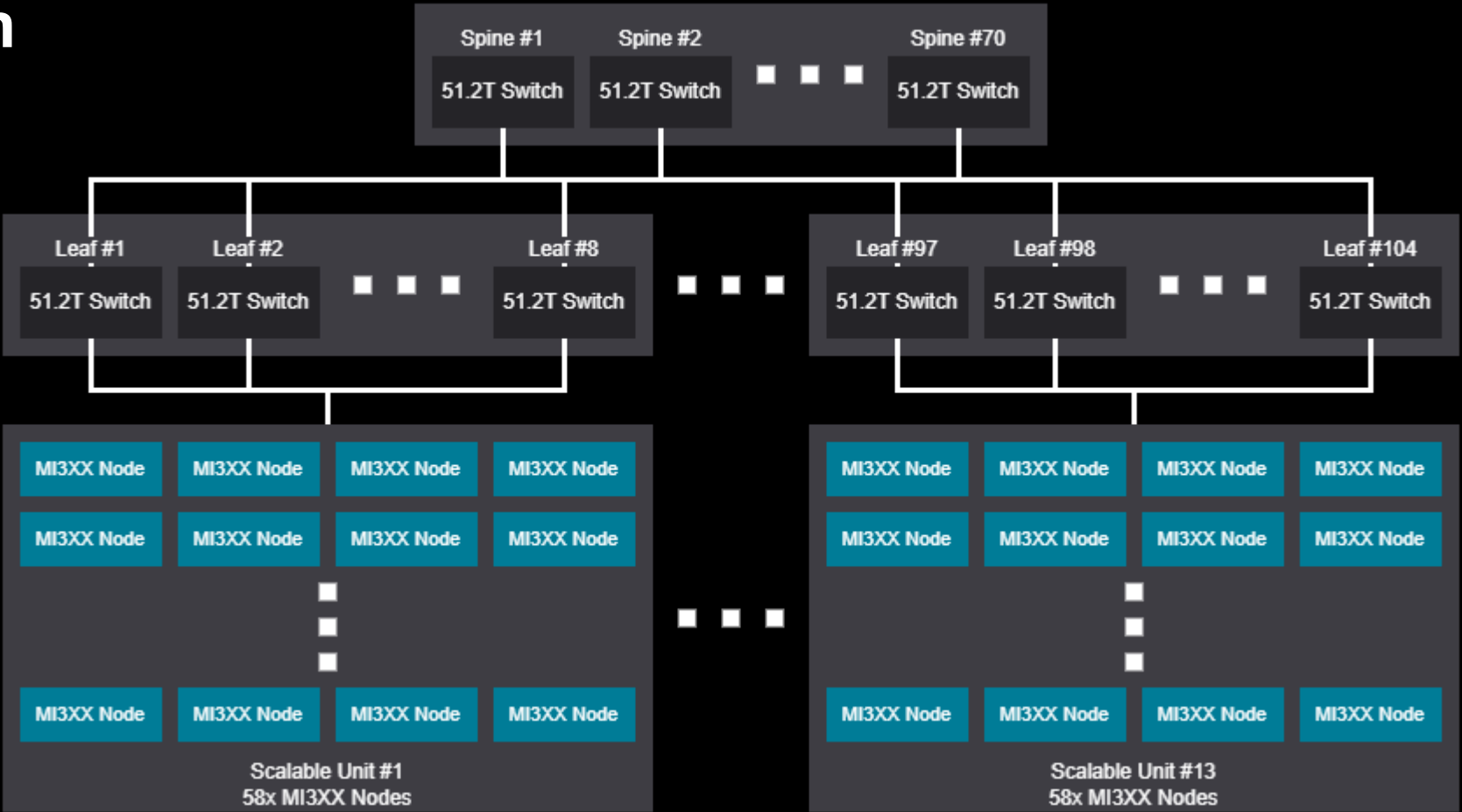
51.2T



Network Diagram - 6032 GPU (754 Nodes)

Rail Design

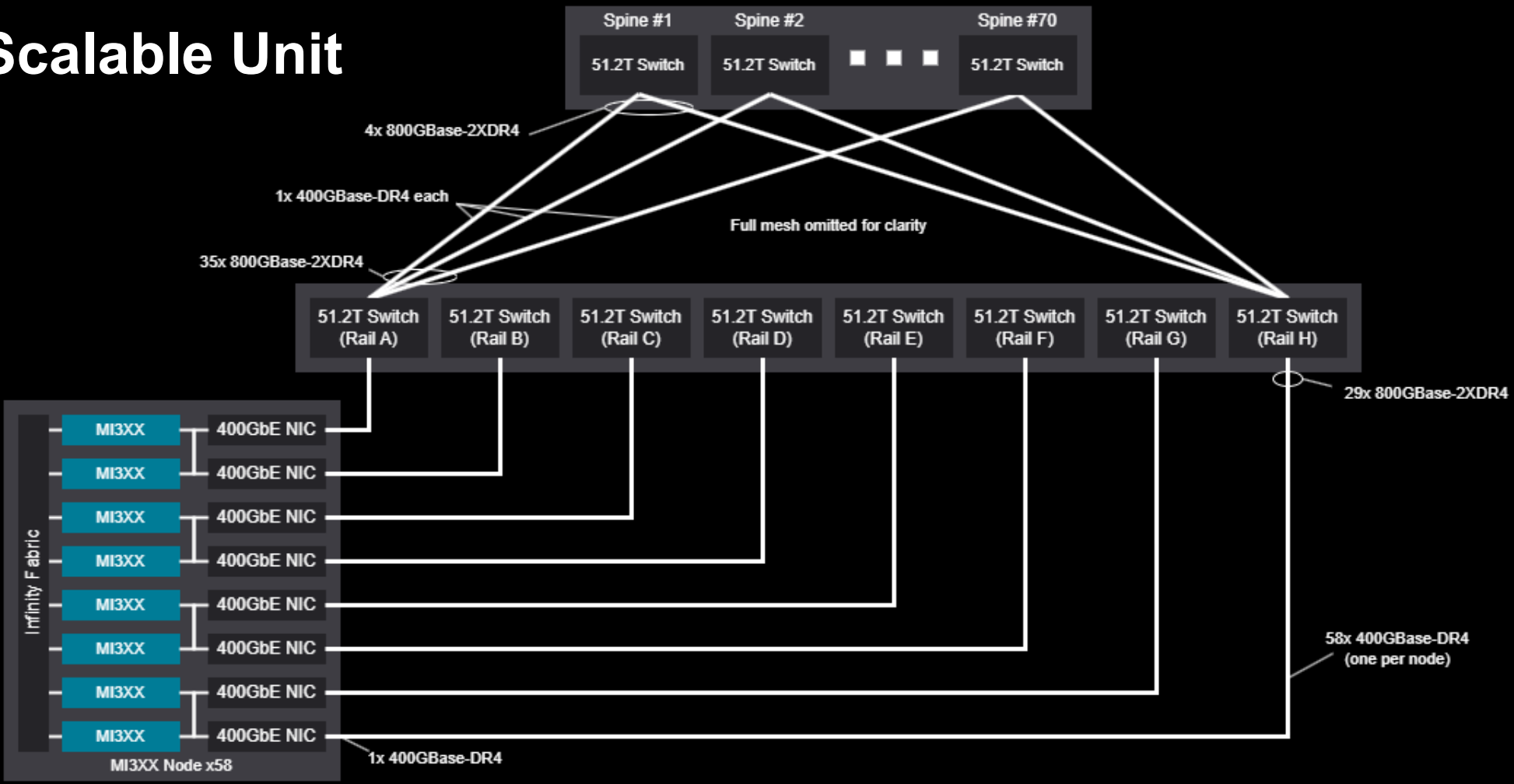
51.2T



Network Diagram - 6032 GPU (754 Nodes)

Rail Scalable Unit

51.2T

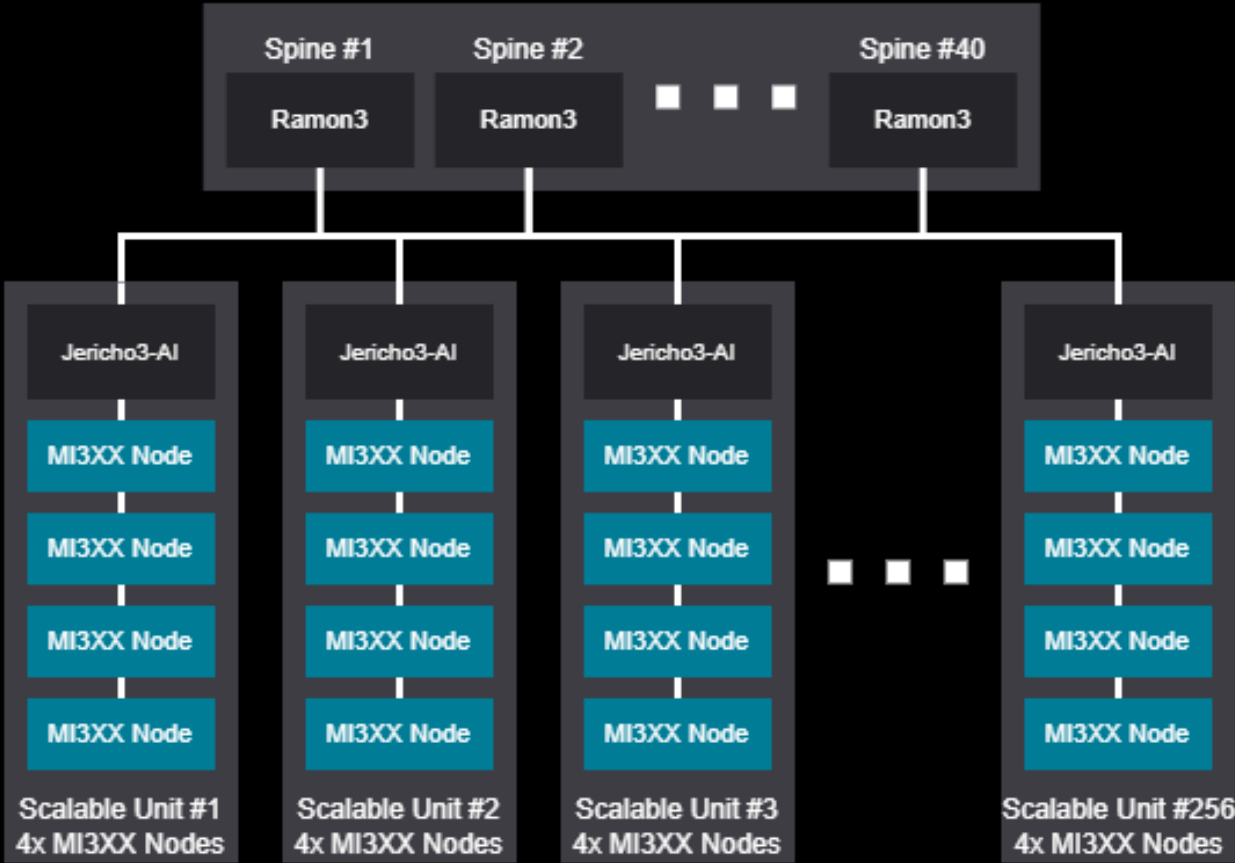


8K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 8192 GPU (1024 Nodes)

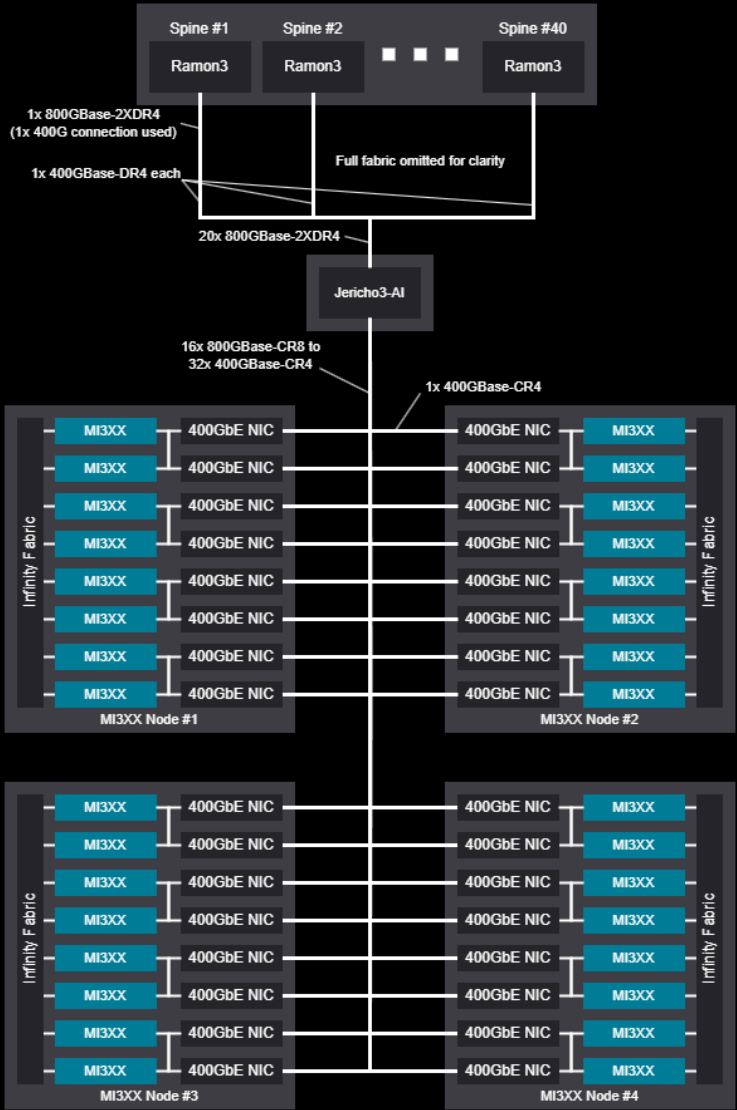
Tree Design

Jericho/Ramon



Network Diagram - 8192 GPU (1024 Nodes)

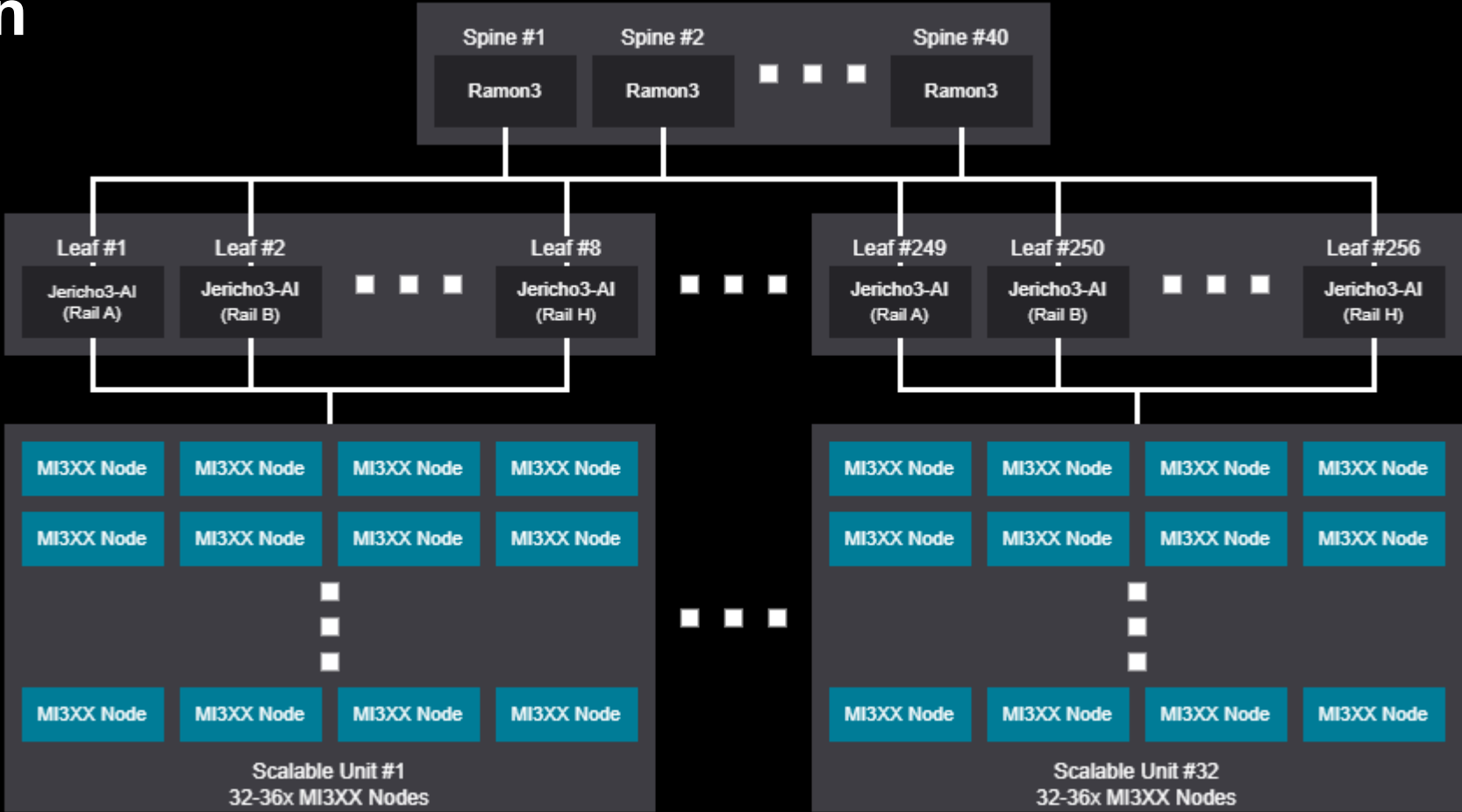
Tree Scalable Unit Jericho/Ramon



Network Diagram - 8192-9216 GPU (1024-1152 Nodes)

Rail Design

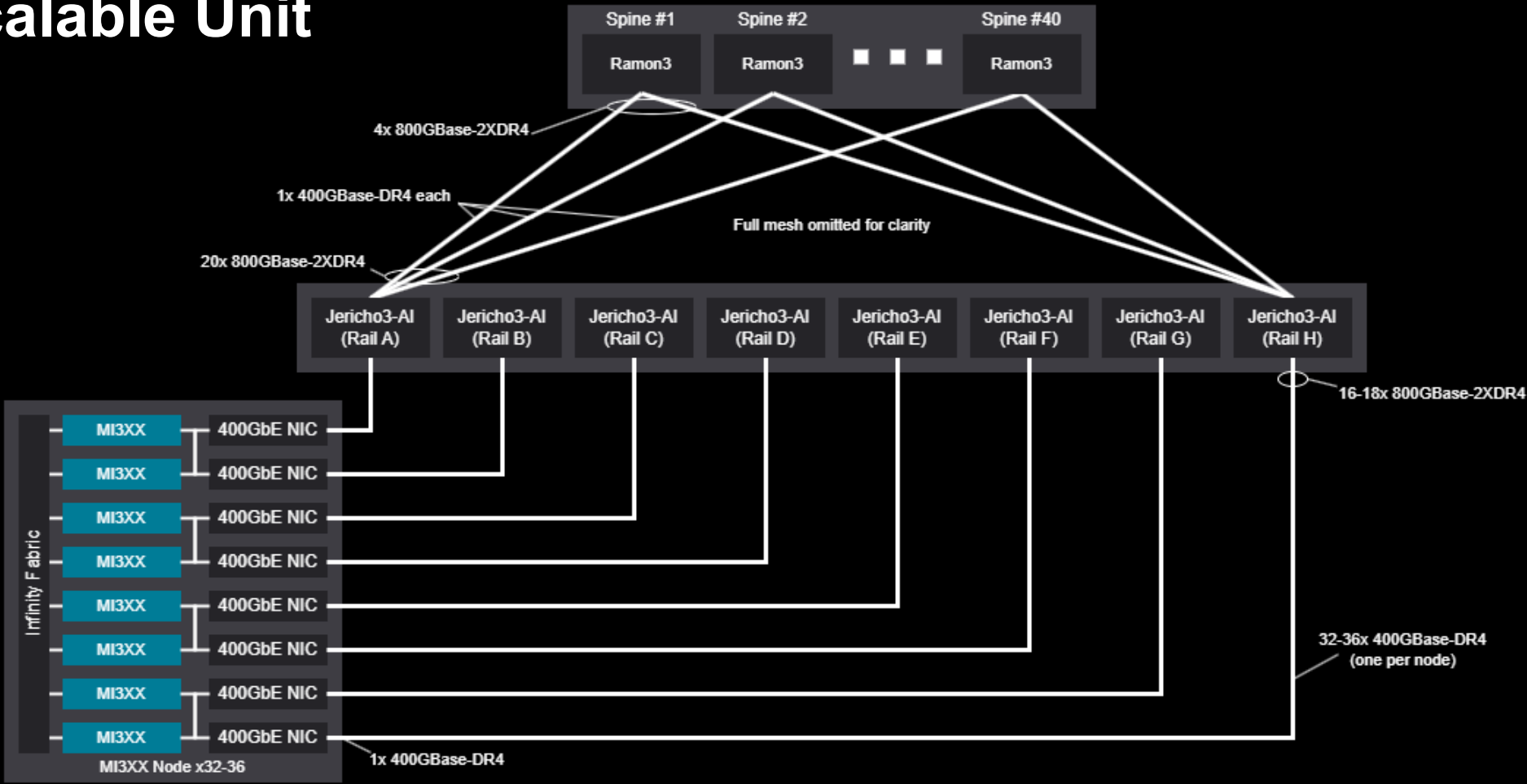
Jericho/Ramon



Network Diagram - 8192-9216 GPU (1024-1152 Nodes)

Rail Scalable Unit

Jericho/Ramon





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